

DISCUSIÓN PROBLEMAS MODELO IDEAL

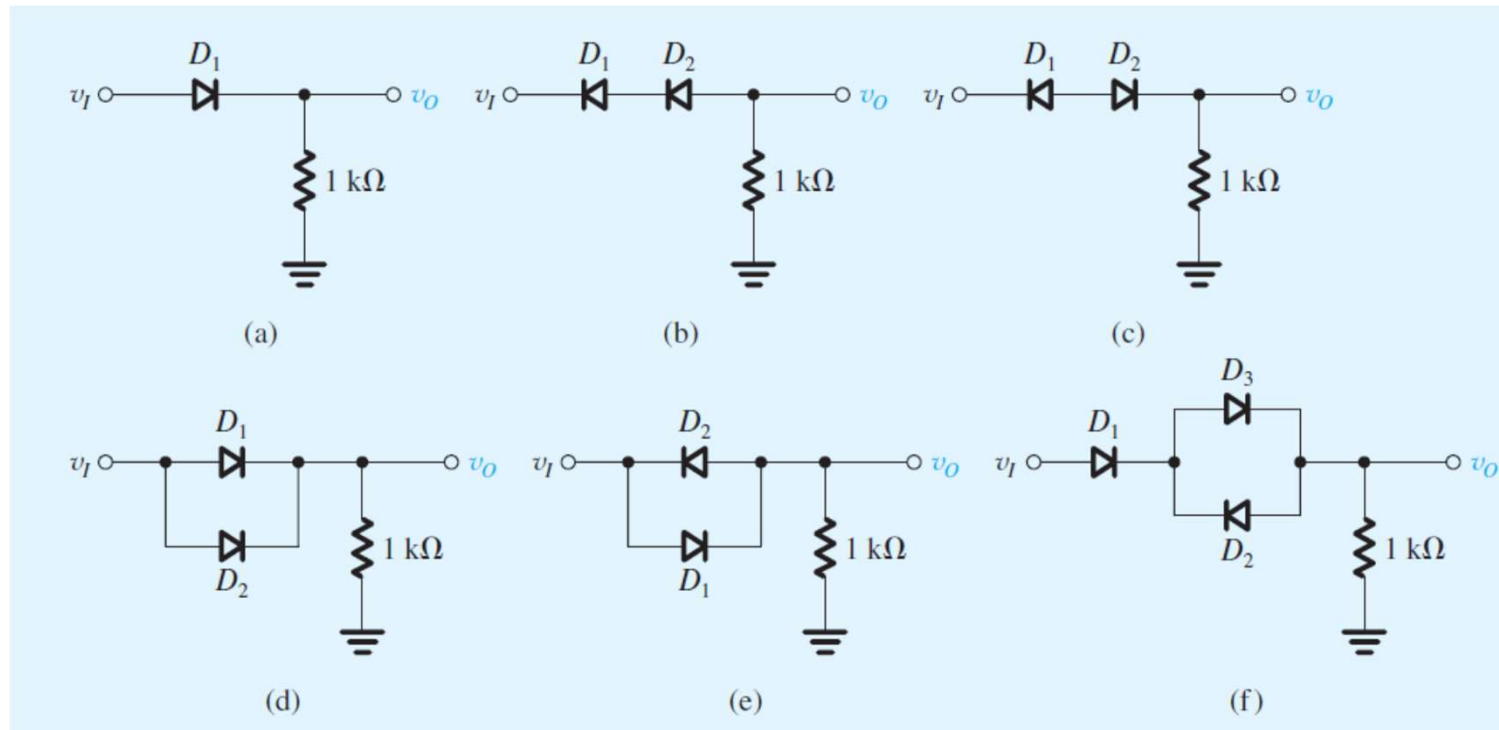
José Ramos López

ELC-115

Escuela de Ingeniería Eléctrica

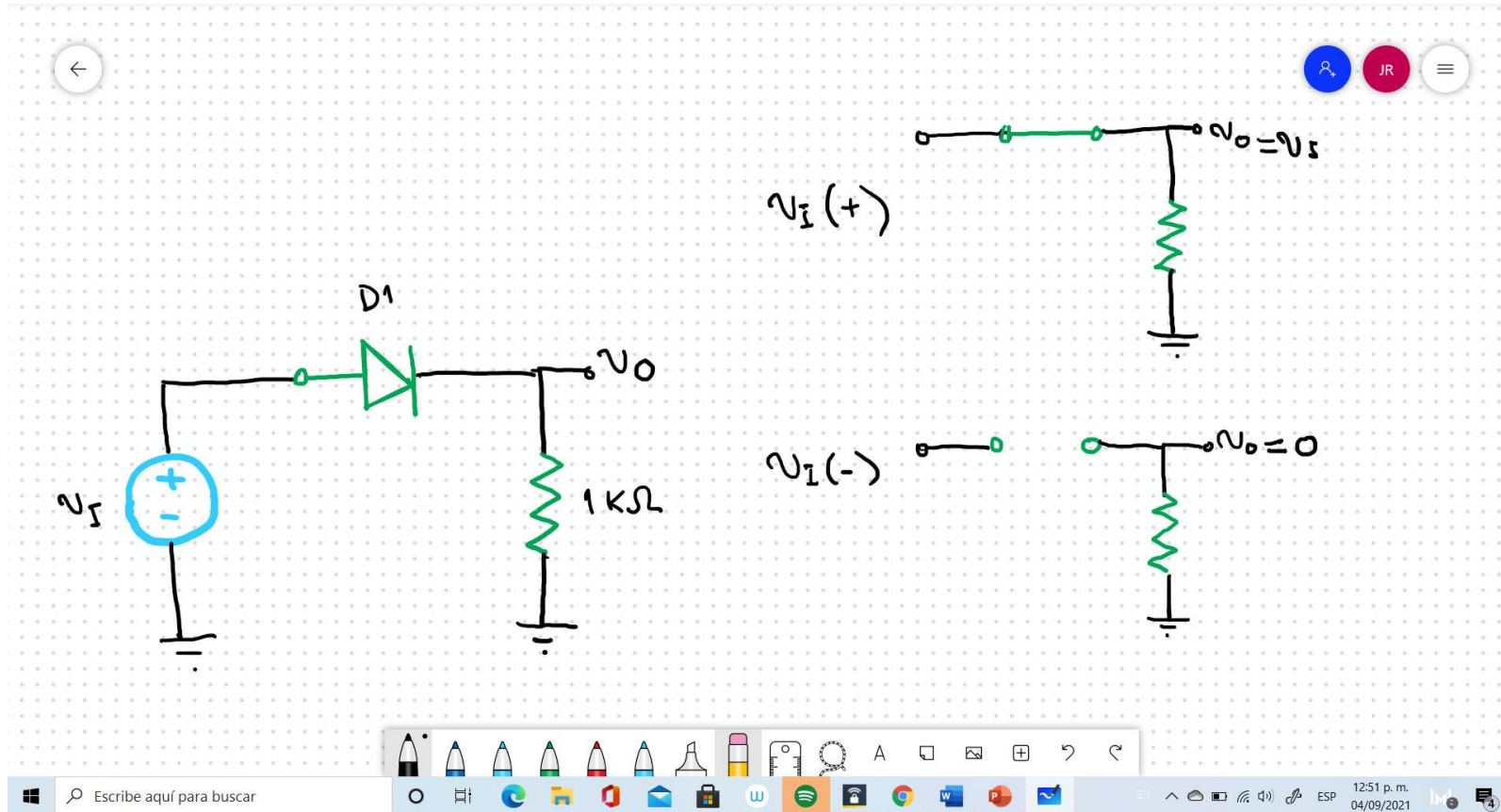
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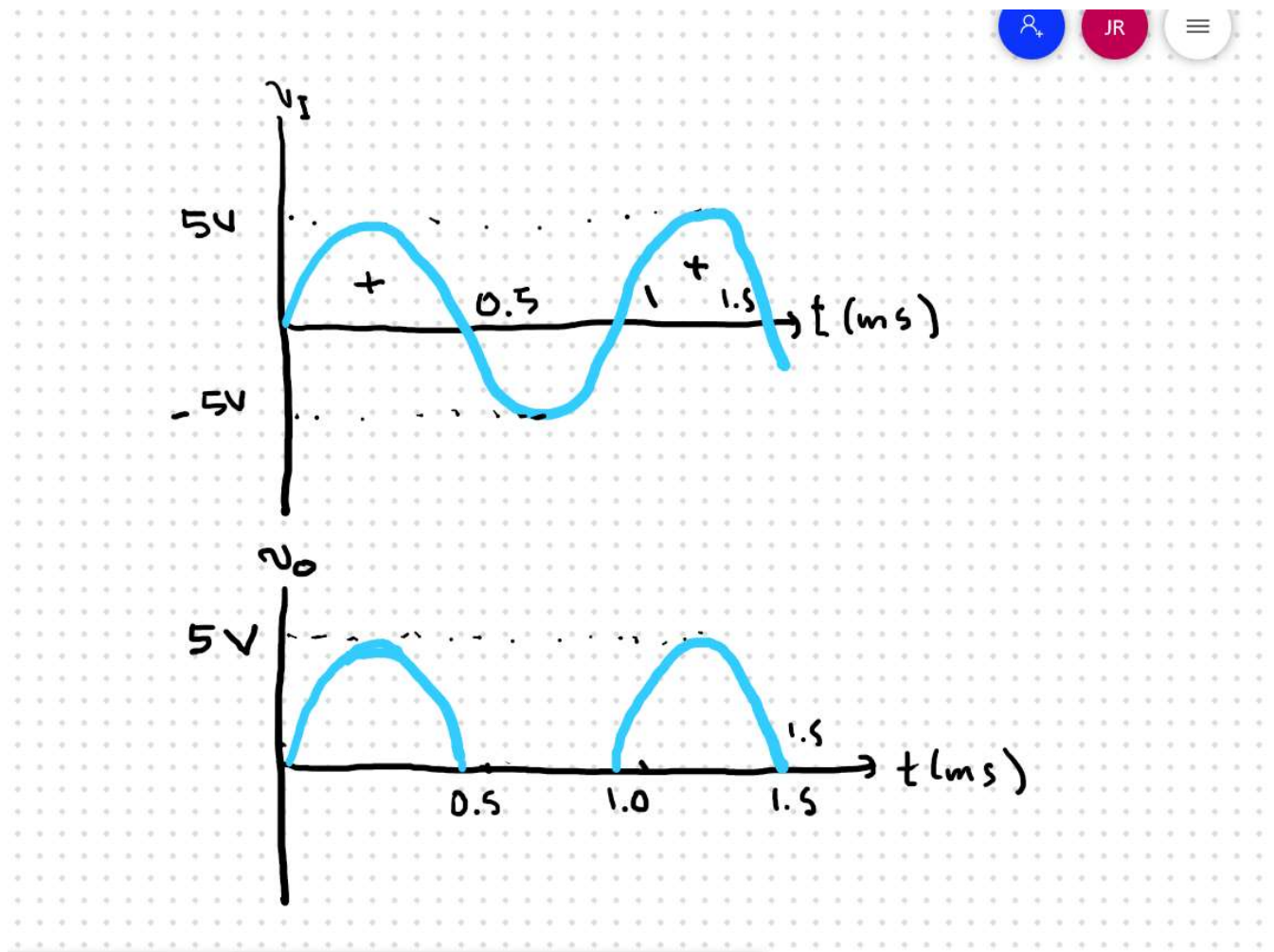
Problema 4. En cada uno de los circuitos, v_I es una onda sinusoidal de 1 kHz con 5V de amplitud. Dibujar v_O

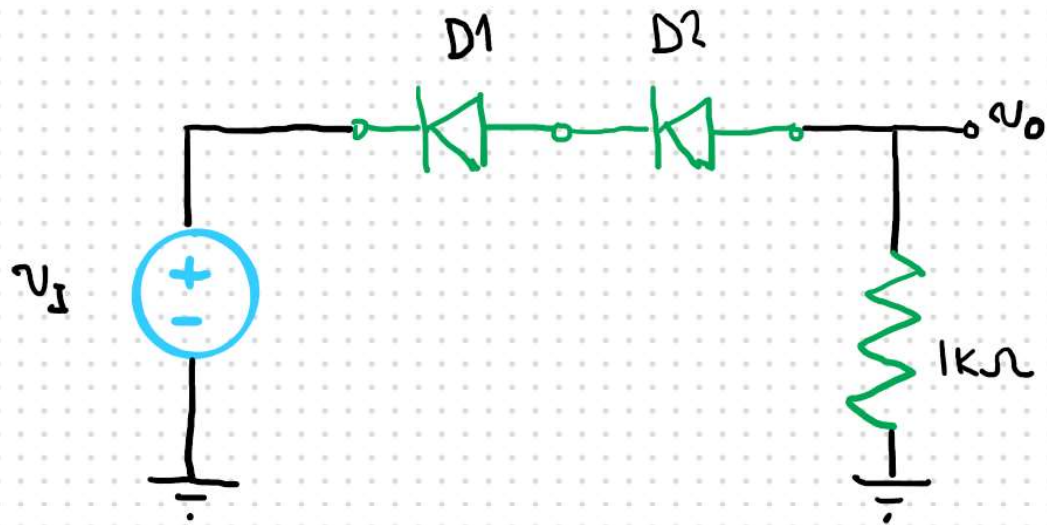


Fuente: Ref. [1]

Problema 4a

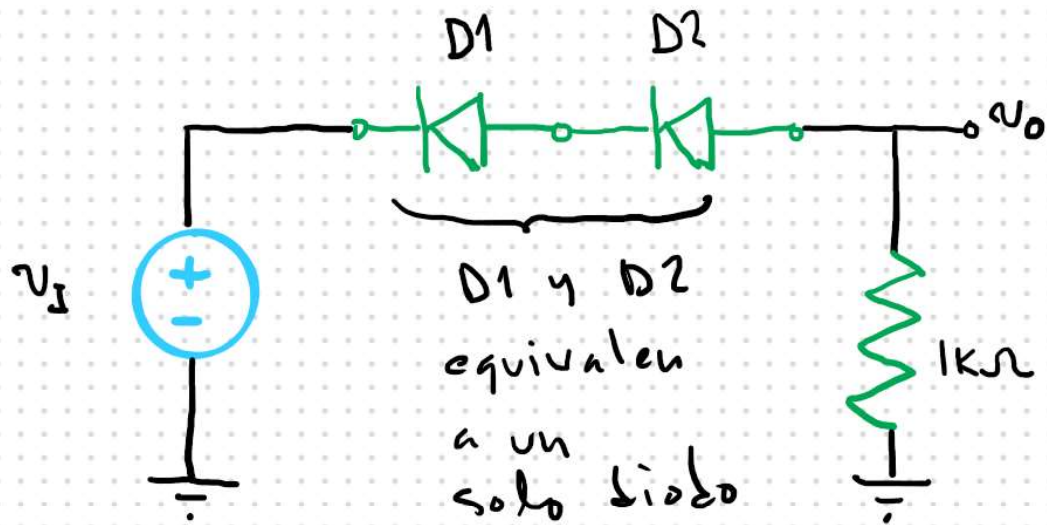






PROBLEMA 4b

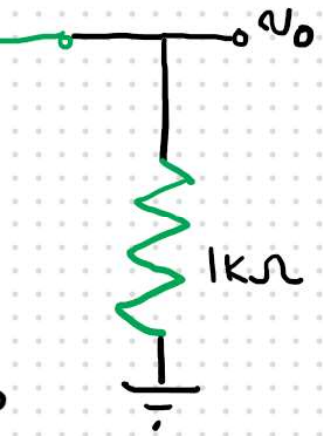
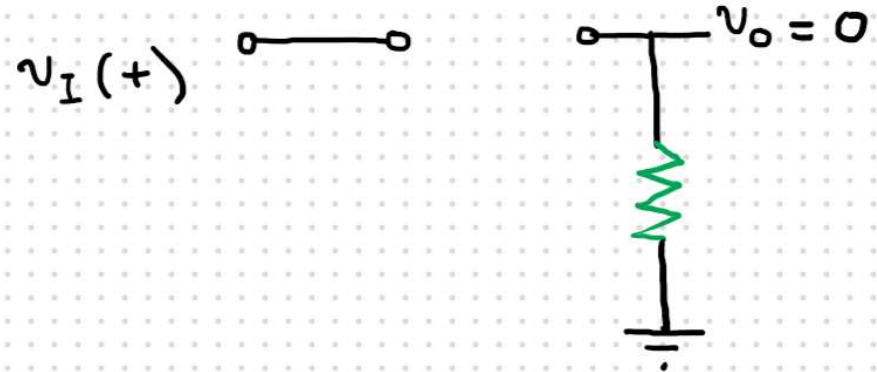
Microsoft Whiteboard toolbar containing drawing tools (pens, highlighters, eraser, selection tools) and a watermark text: "Universidad de El Salvador 2021".



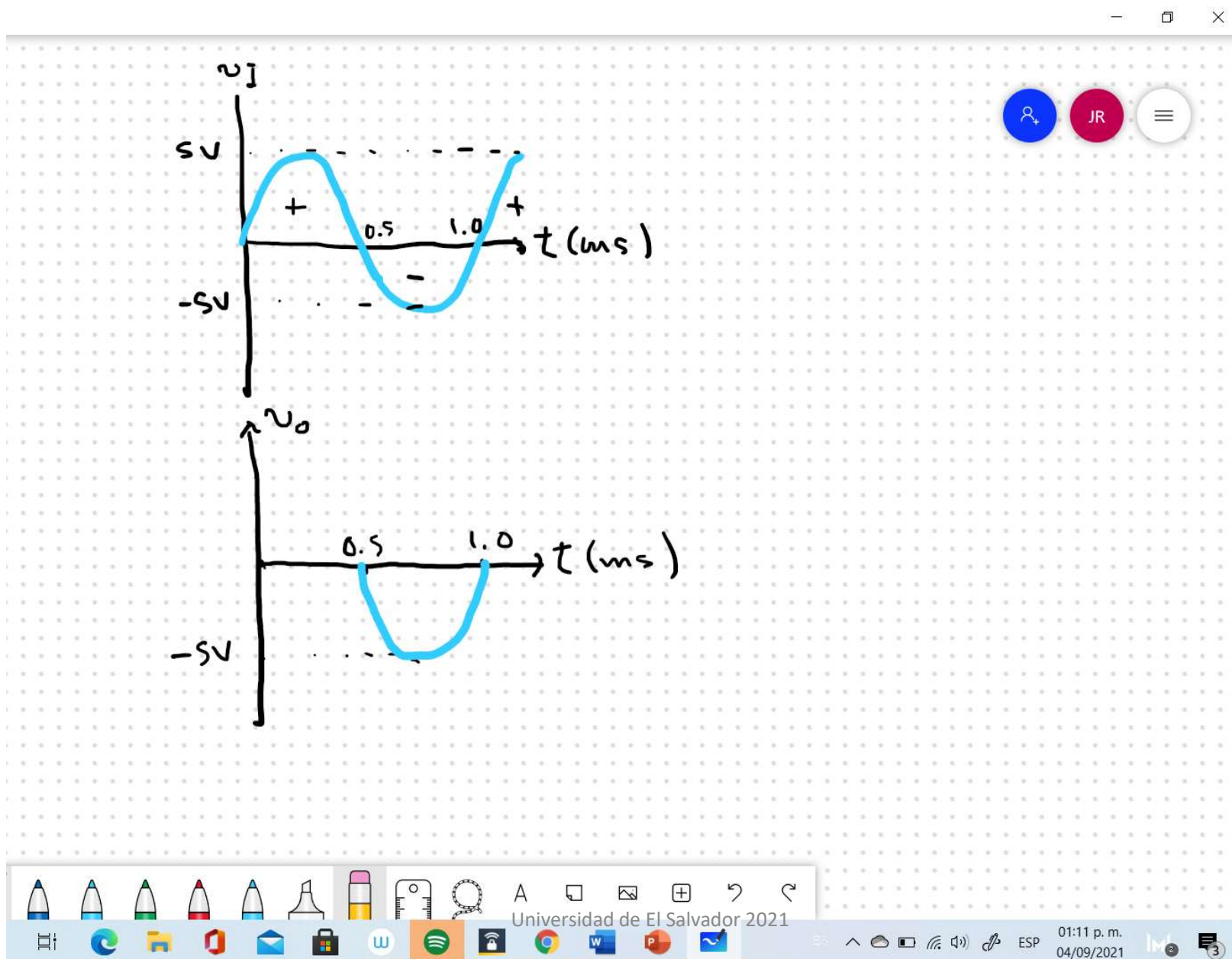
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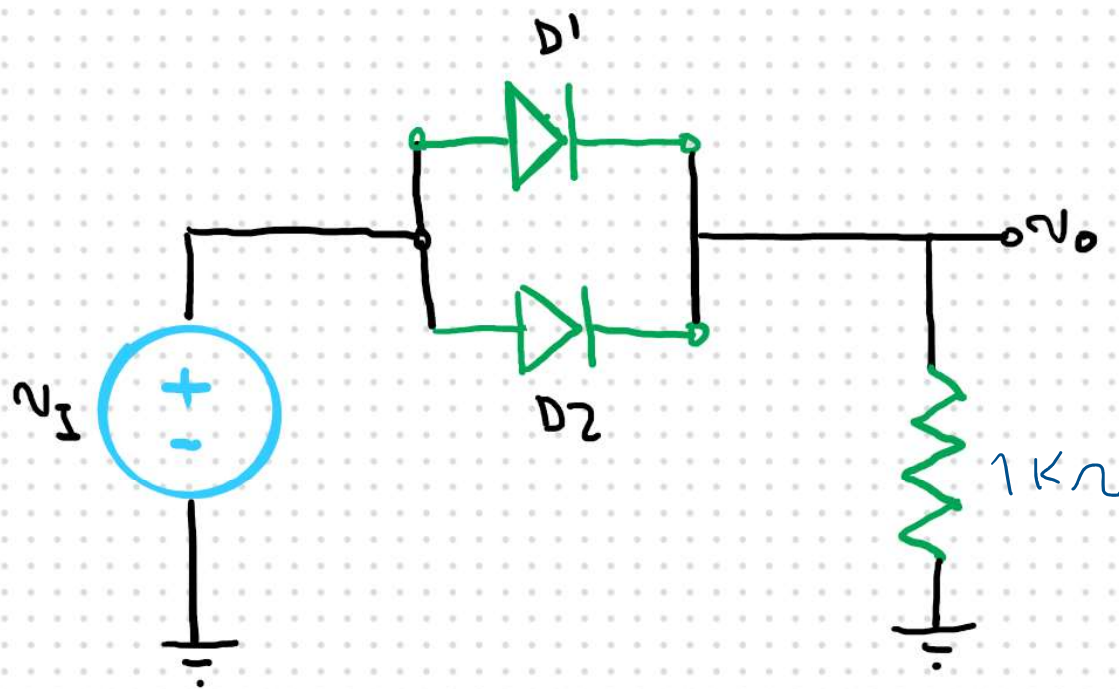


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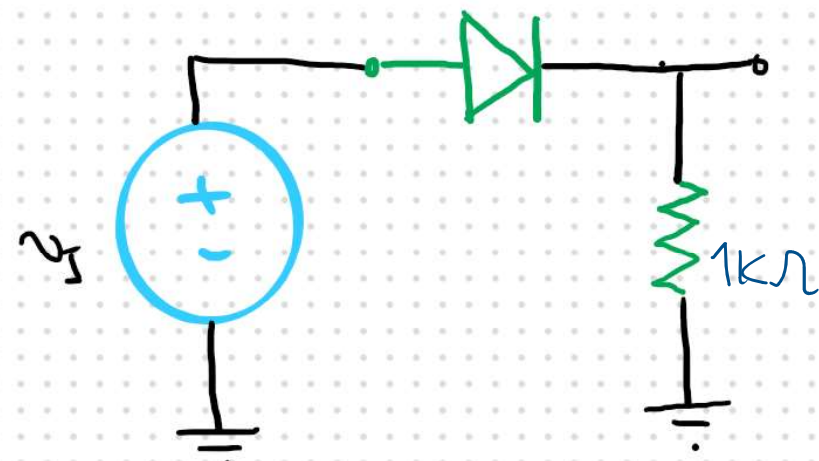
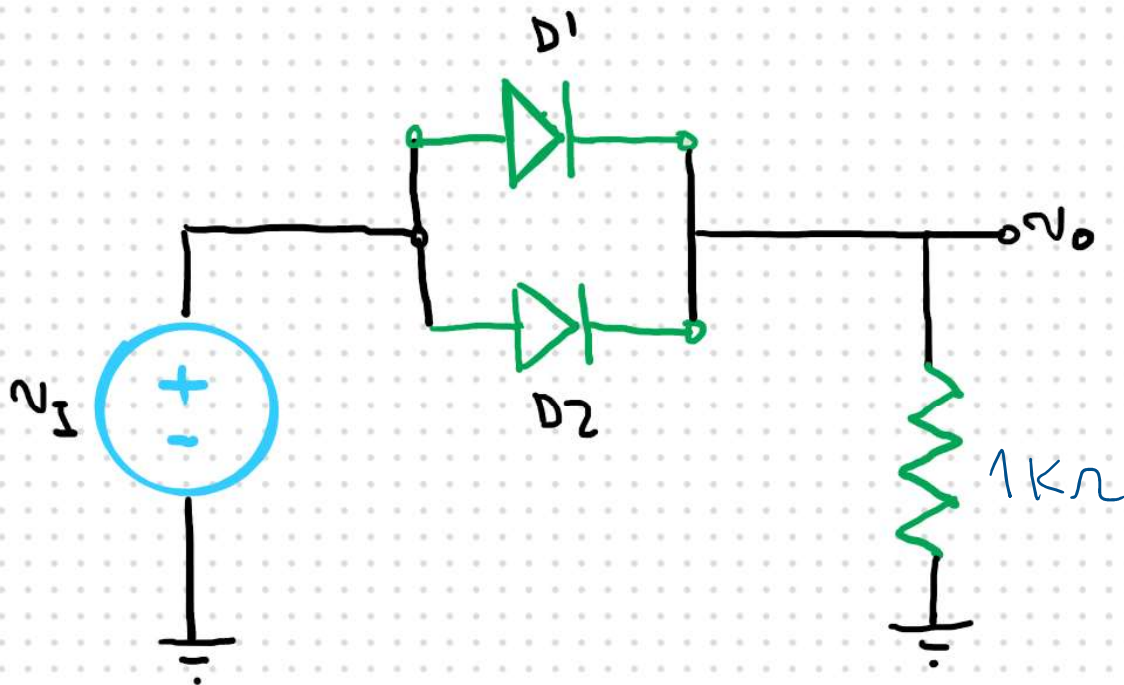
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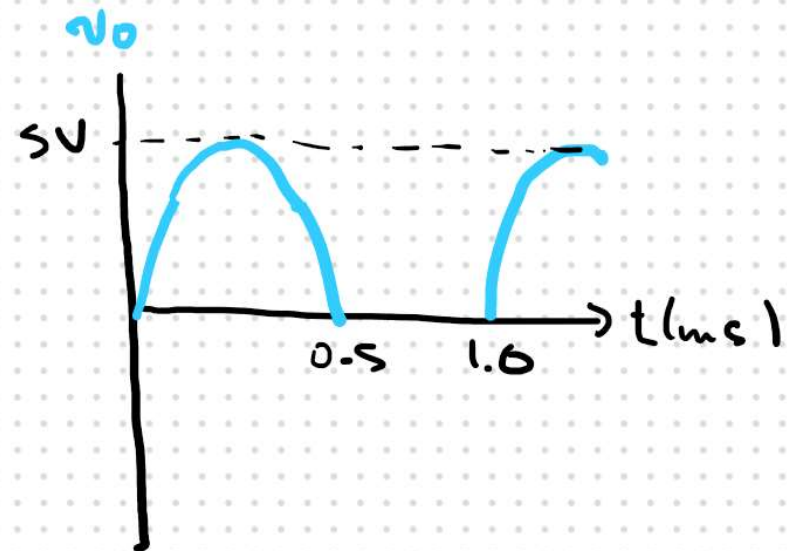
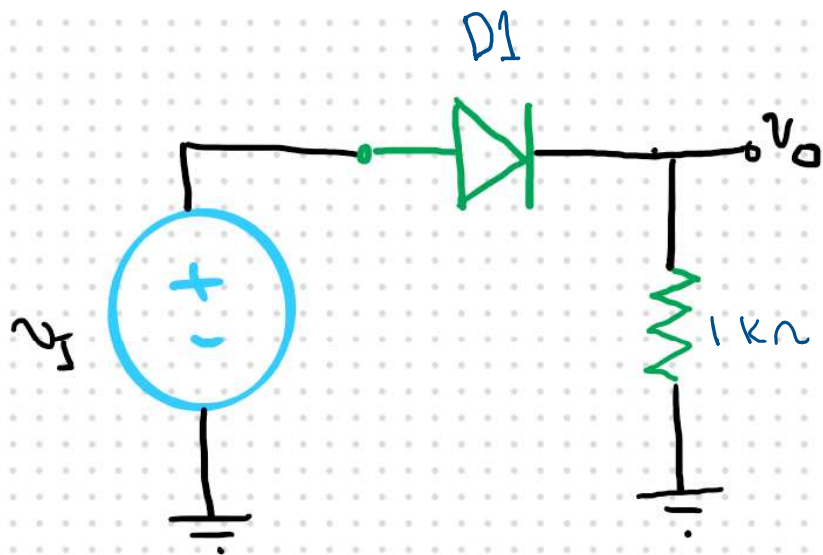
(d)

Microsoft Whiteboard toolbar containing various drawing tools such as pens, highlighters, eraser, and selection tools. A watermark "Universidad de El Salvador 2021" is visible in the background of the toolbar.

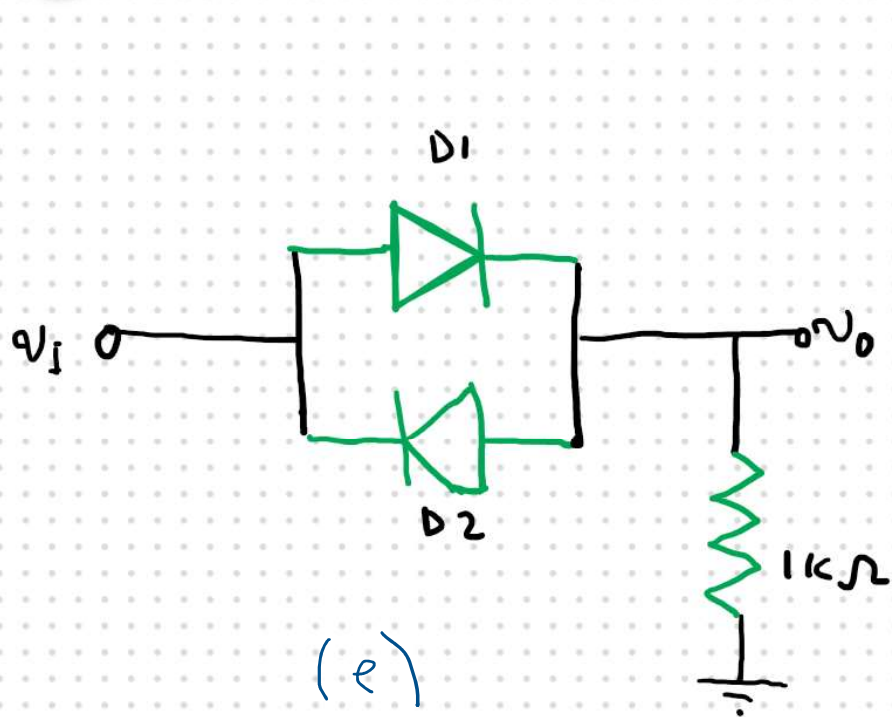


(a)

D1 y D2 equivalen a un solo diodo

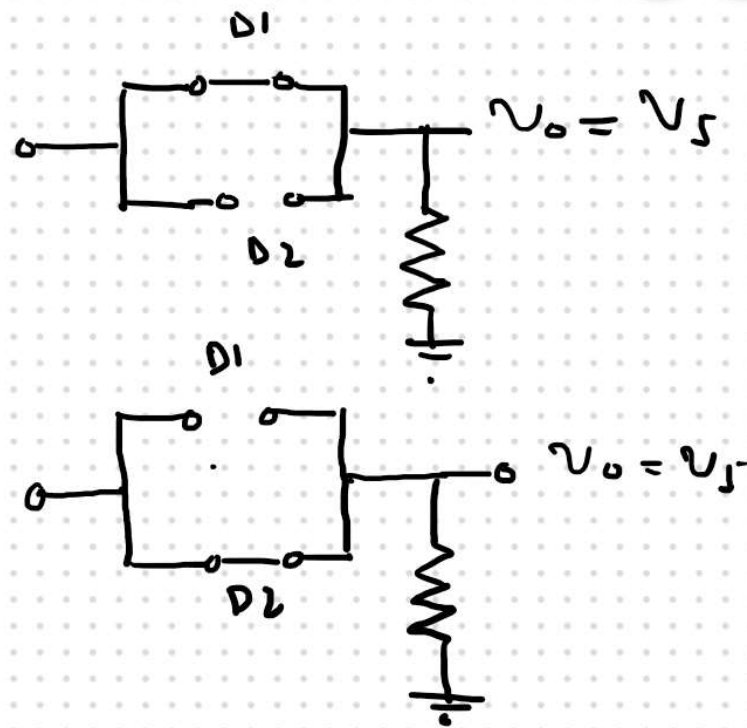


Por lo tanto, la salida es la misma que la del literal (a)



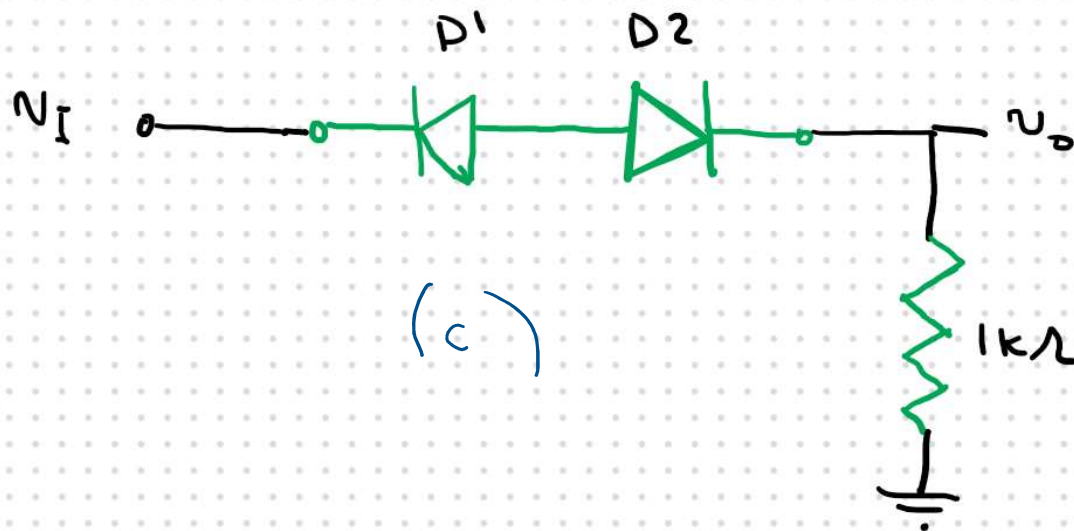
$v_i(+)$

$v_i(-)$



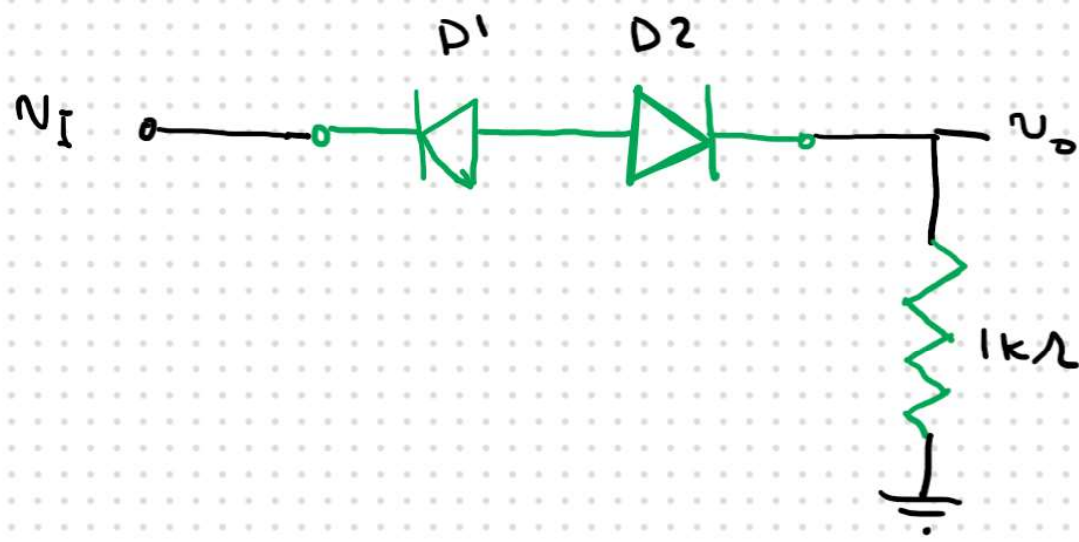


JR

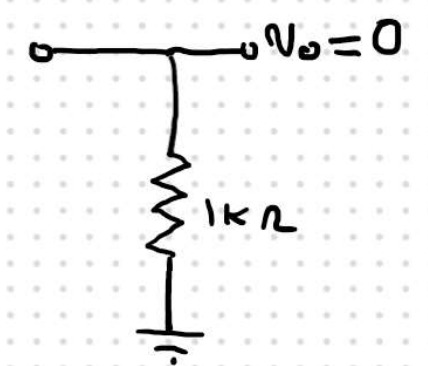


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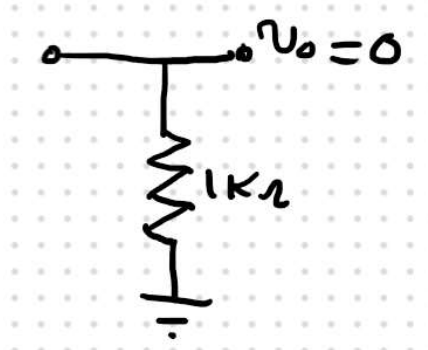
Microsoft Whiteboard toolbar with drawing tools (pen, highlighter, eraser, selection tools) and application icons (Edge, File Explorer, Office apps, etc.).



$v_I(+)$

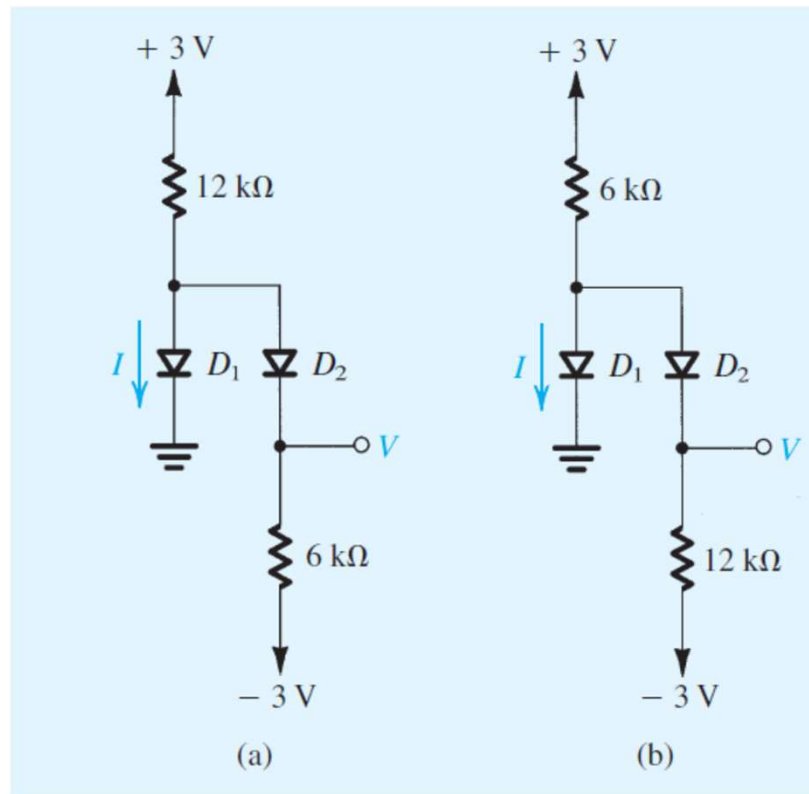


$v_I(-)$

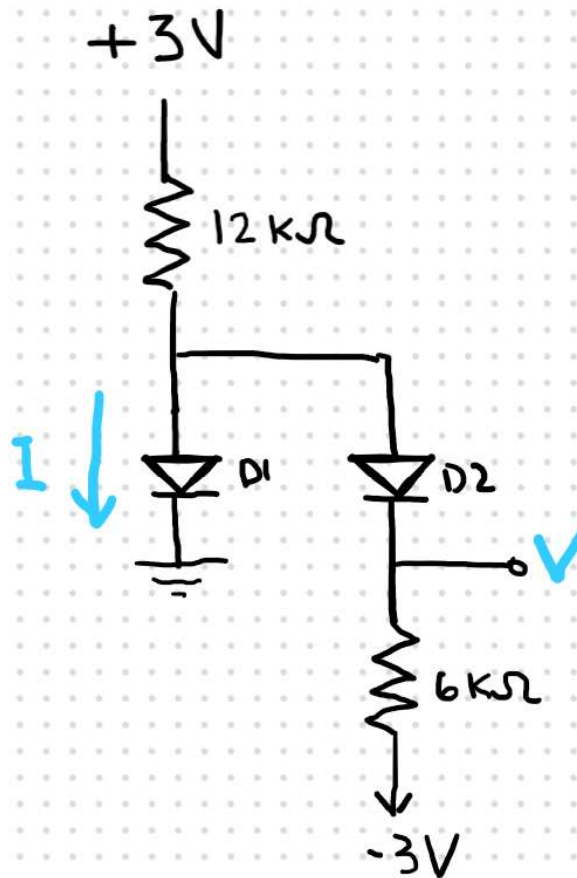


Microsoft Whiteboard toolbar with drawing tools (pen, highlighter, eraser, selection tools) and a watermark: "Universidad de El Salvador 2021".

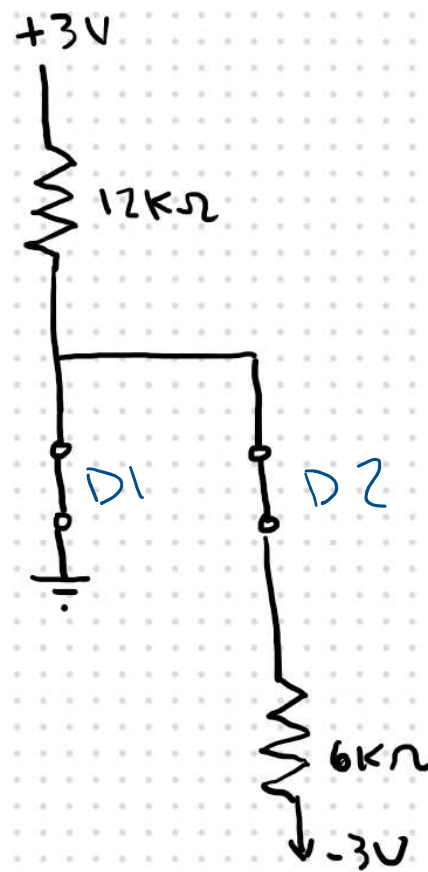
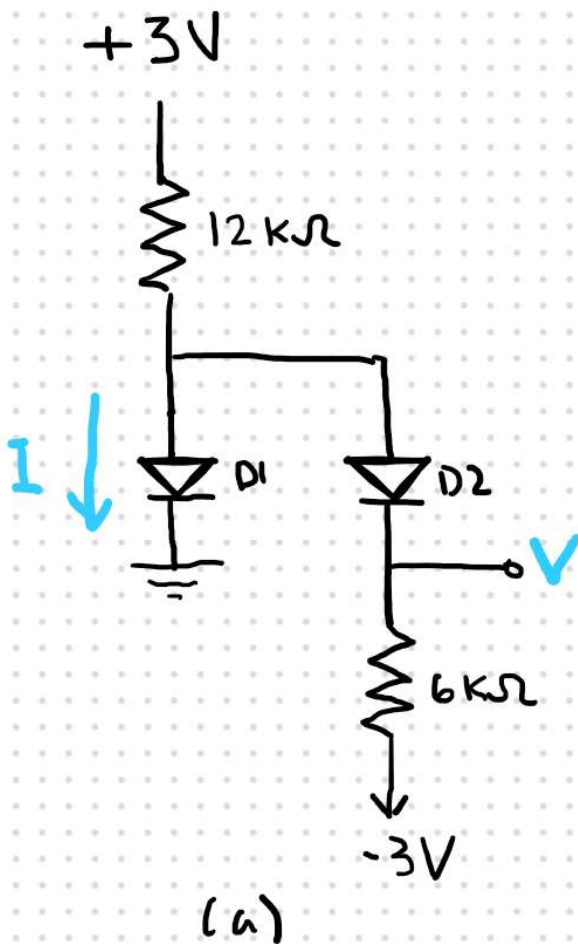
Problema 9. Para diodos ideales, encontrar las tensiones y corrientes indicadas

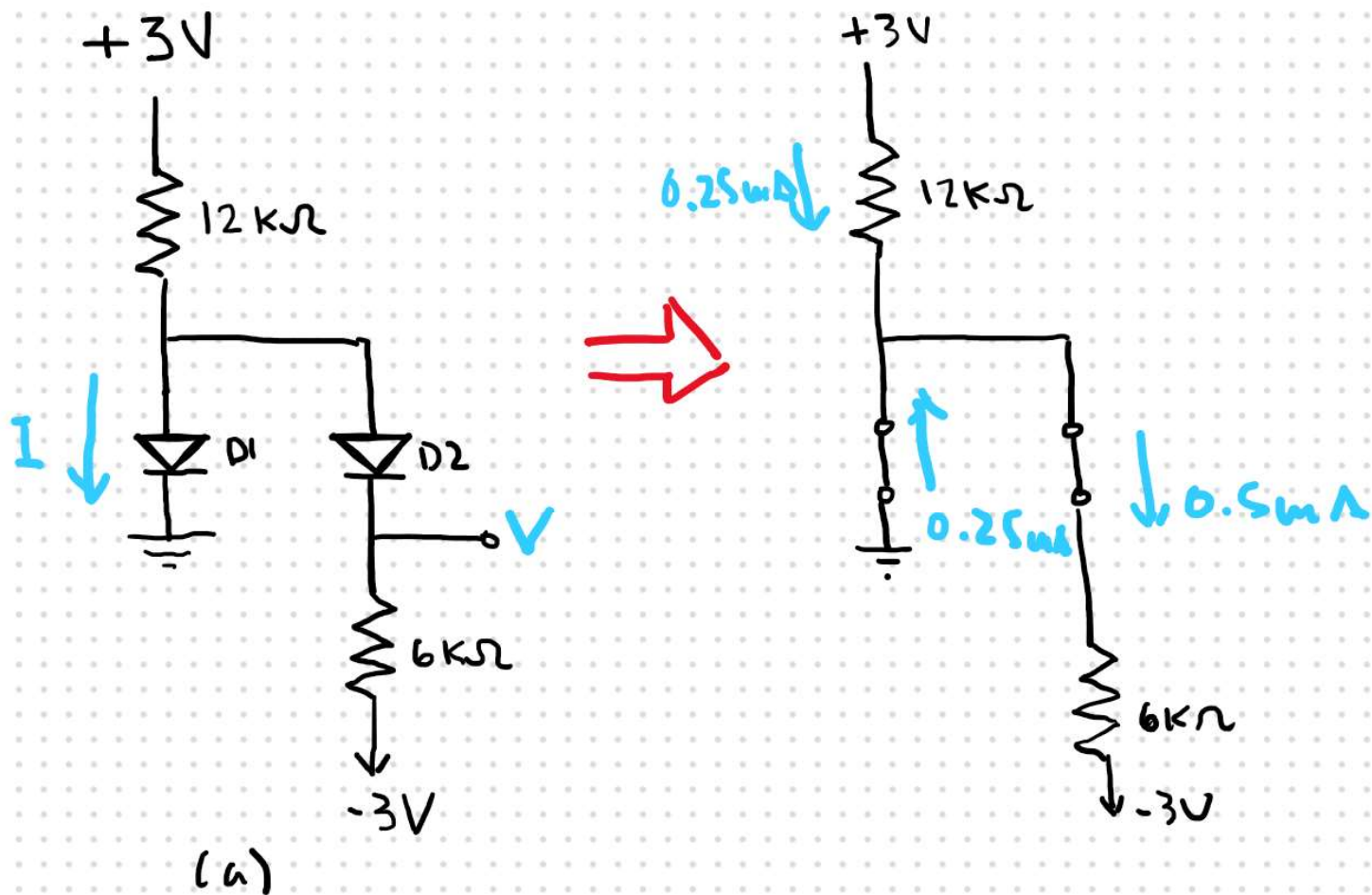


Fuente: Ref. [1]



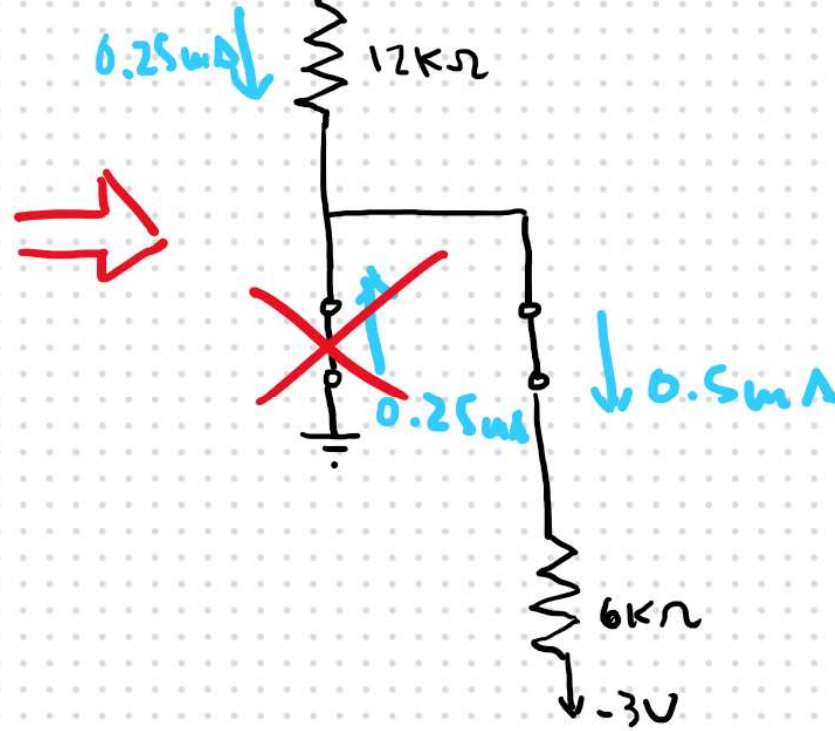
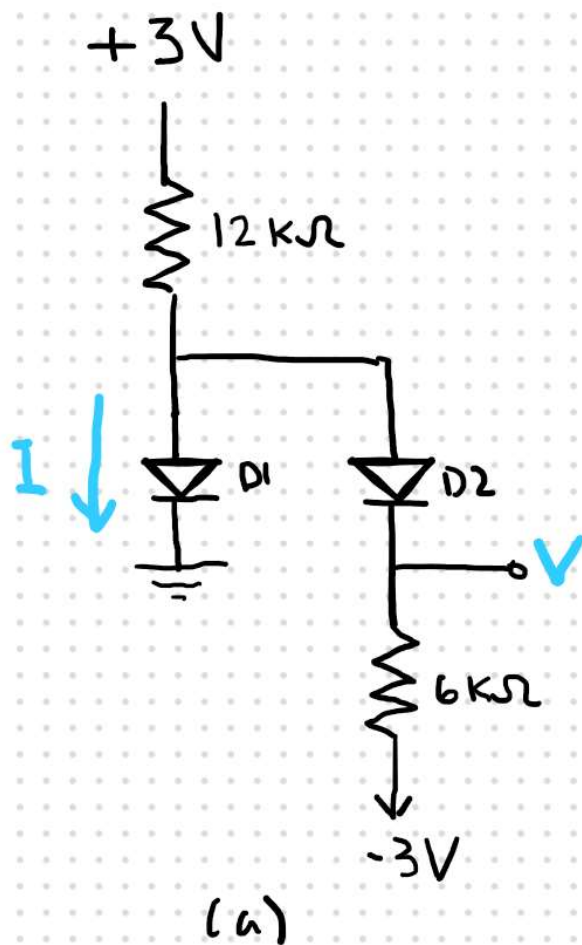
(a)

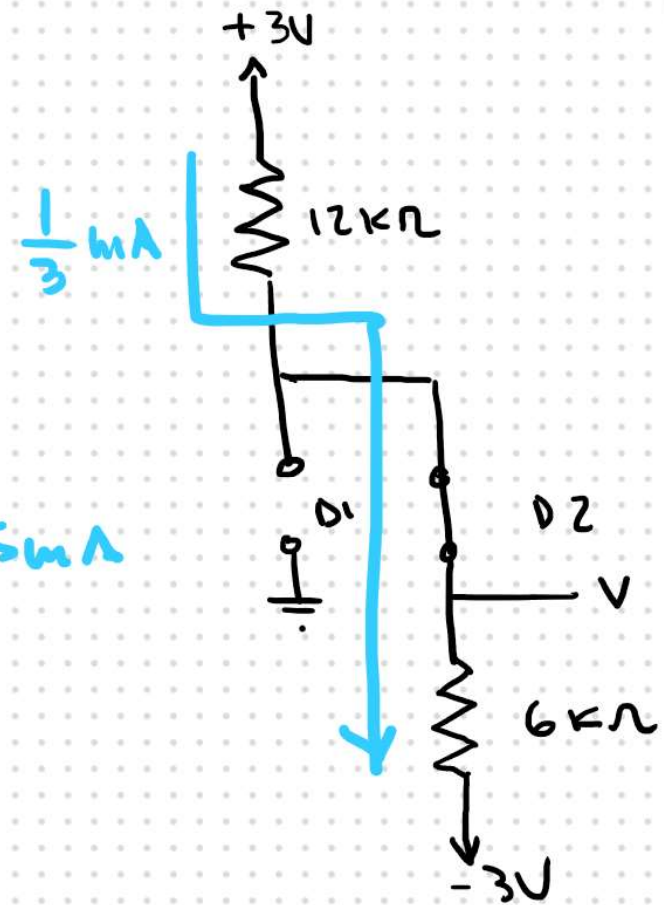
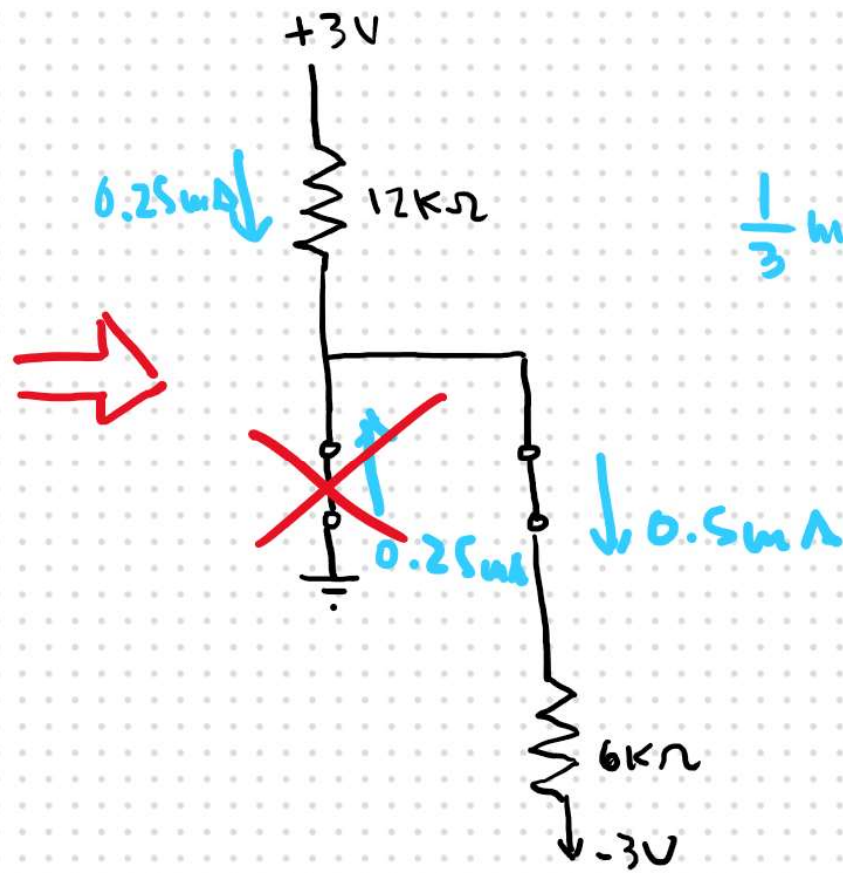
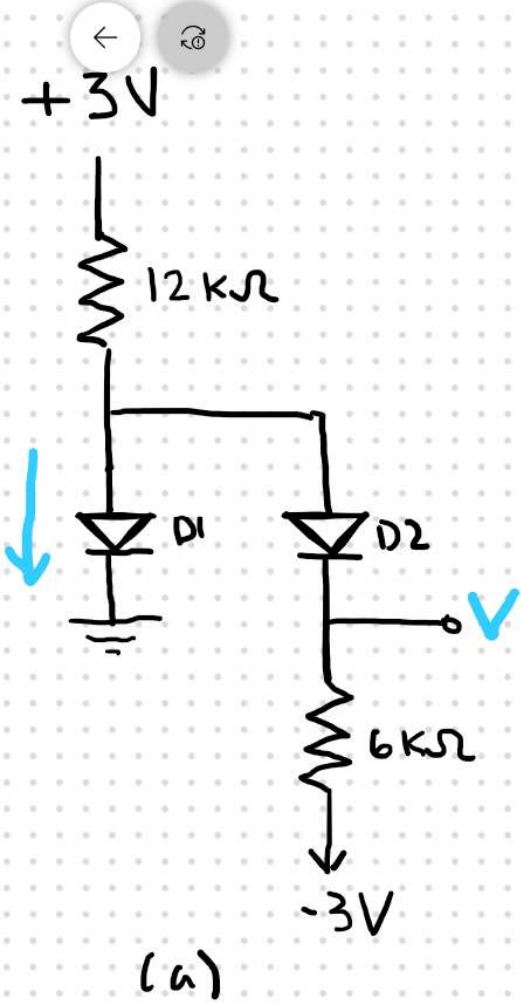


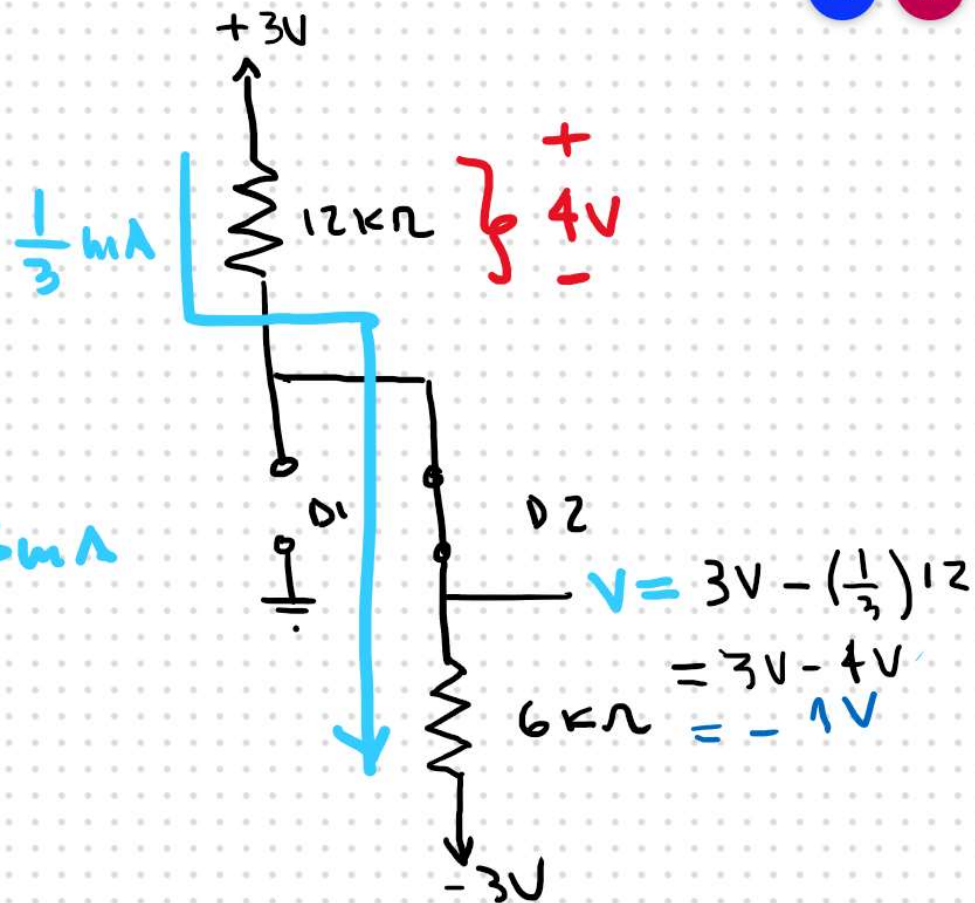
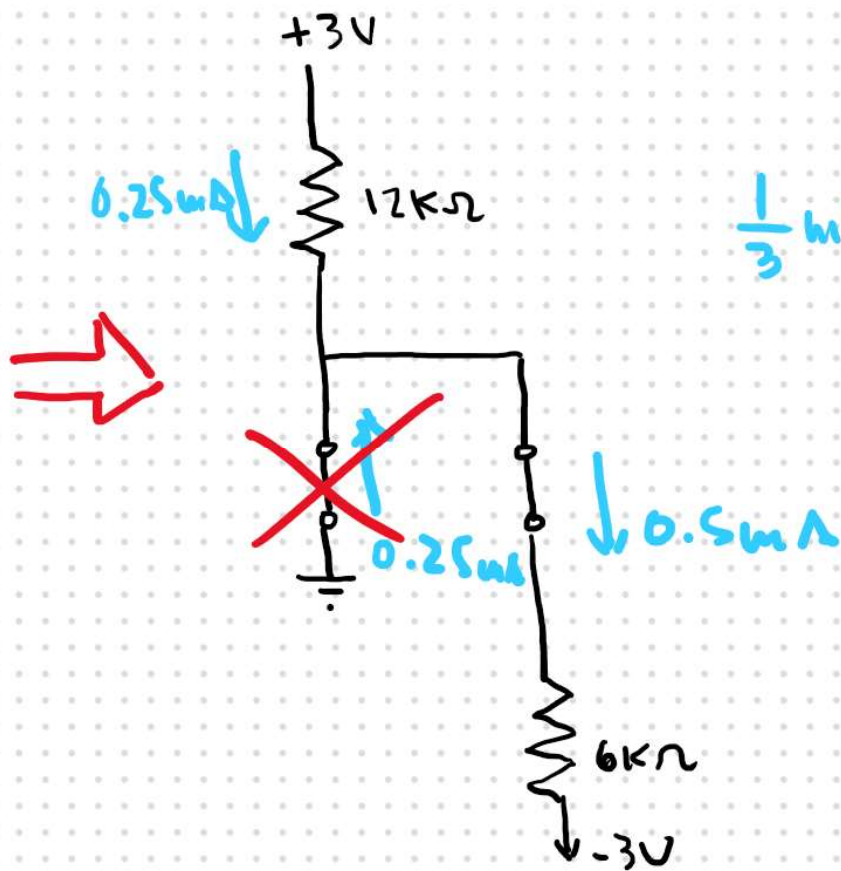
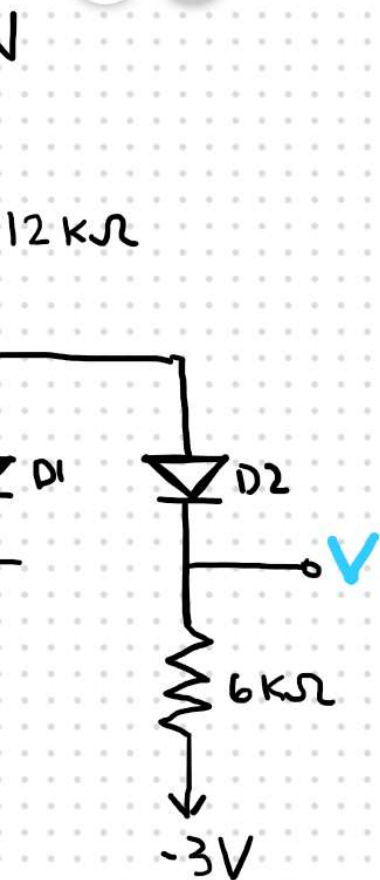


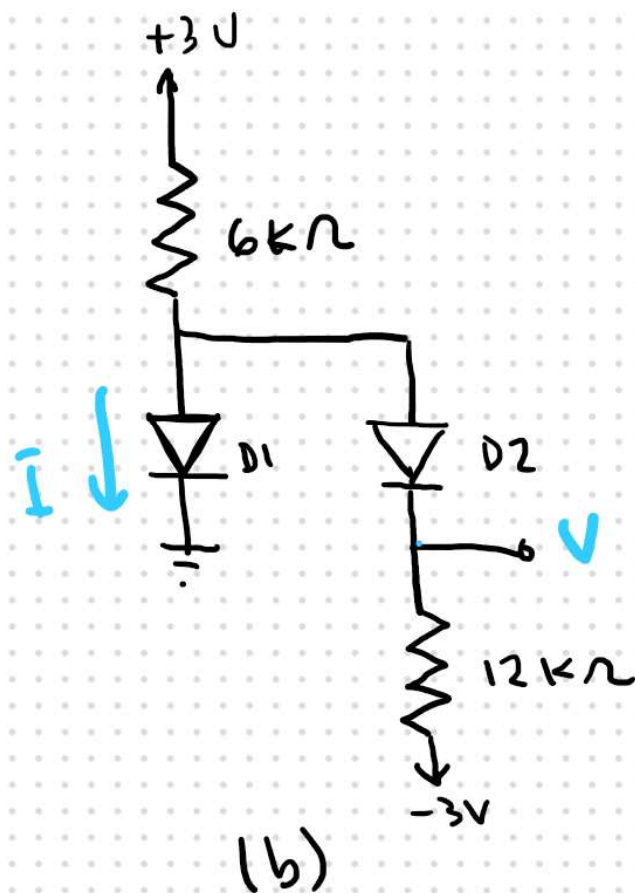
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Tools: Eraser, Highlighter, Pencil, Pen, Lasso, Text, Image, Shape, Line, Arrow, Eraser, Highlighter, Pencil, Pen, Lasso, Text, Image, Shape, Line, Arrow



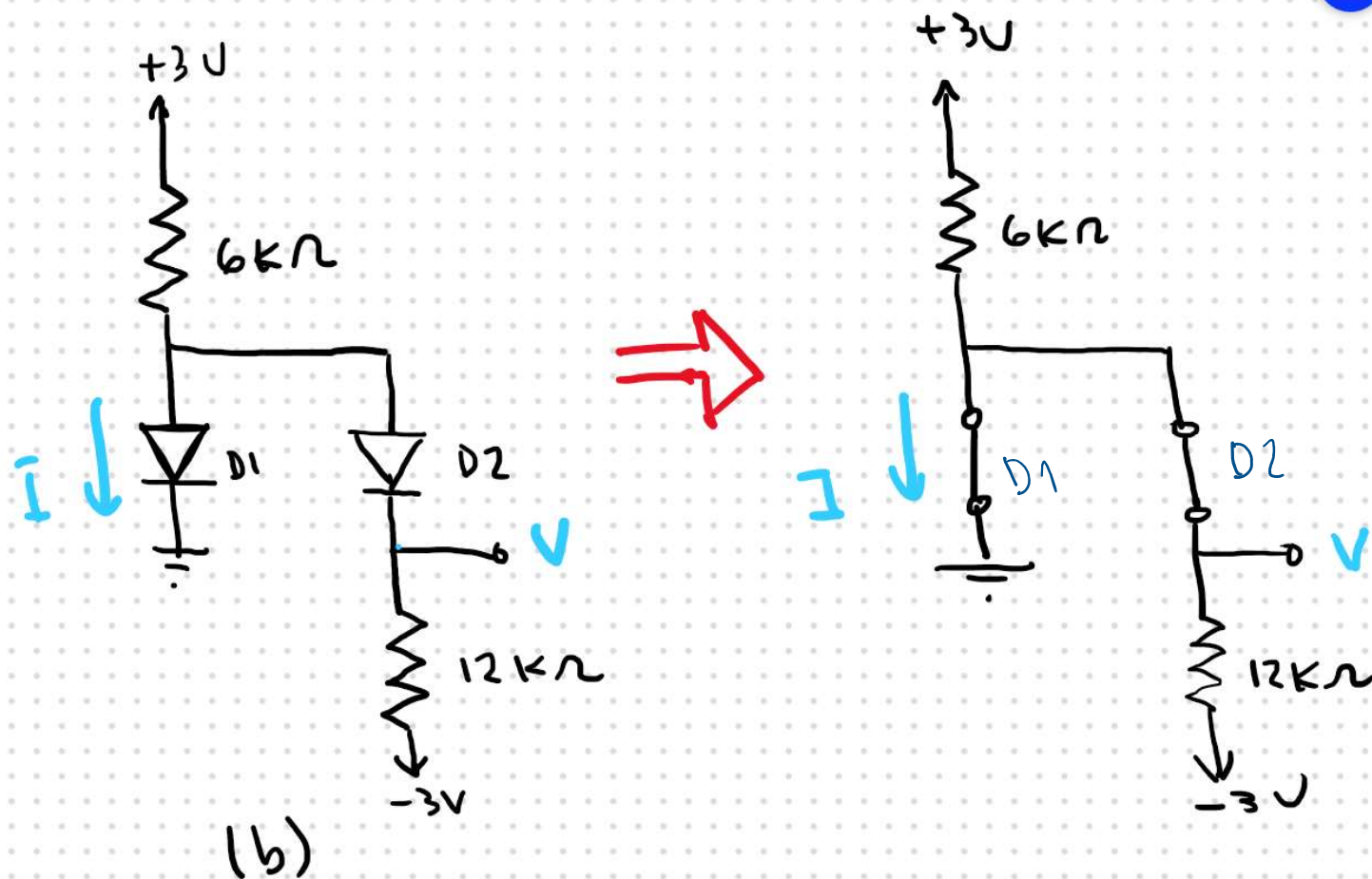


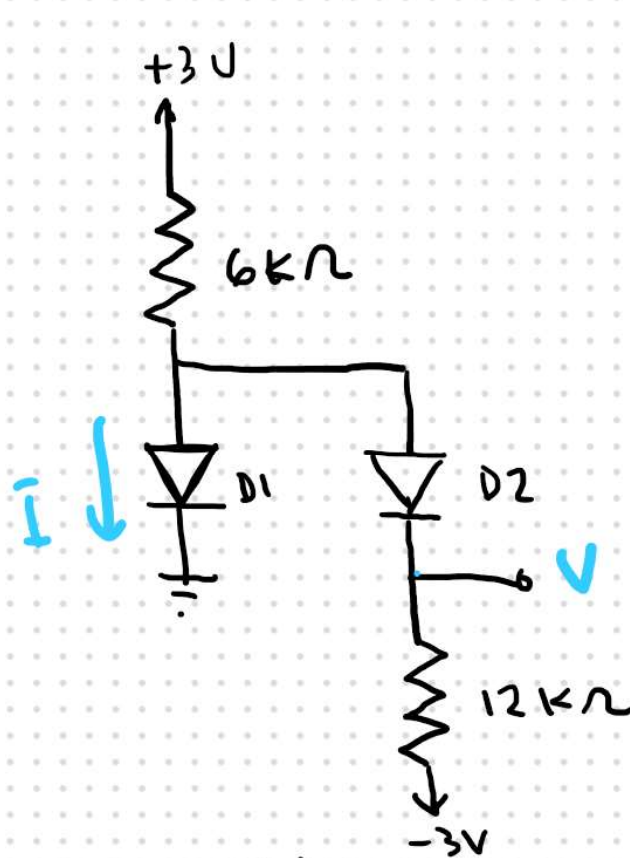




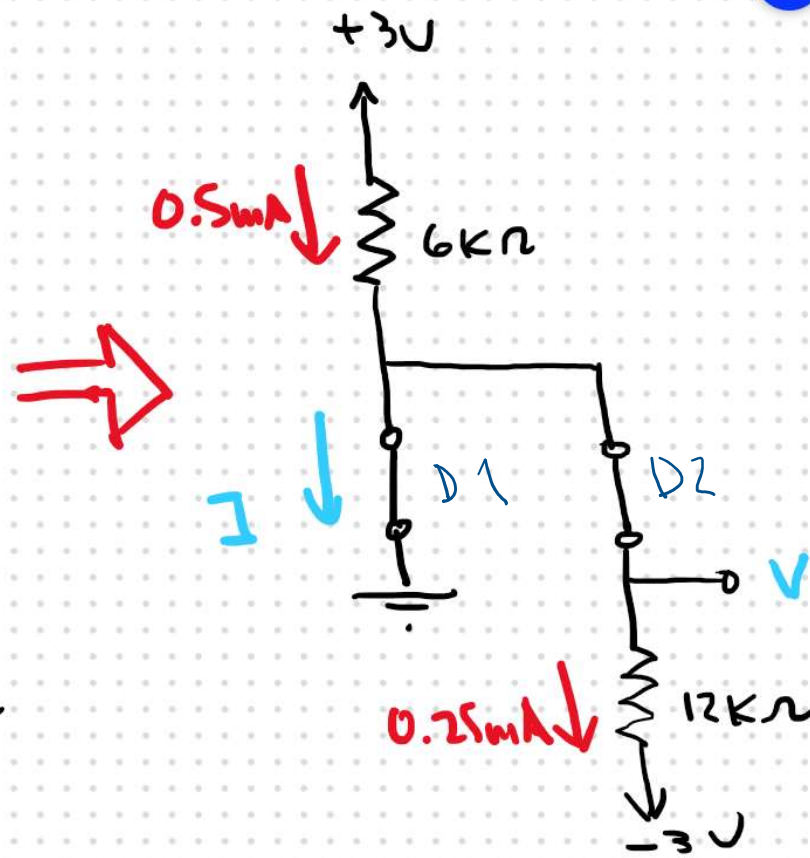
(b)

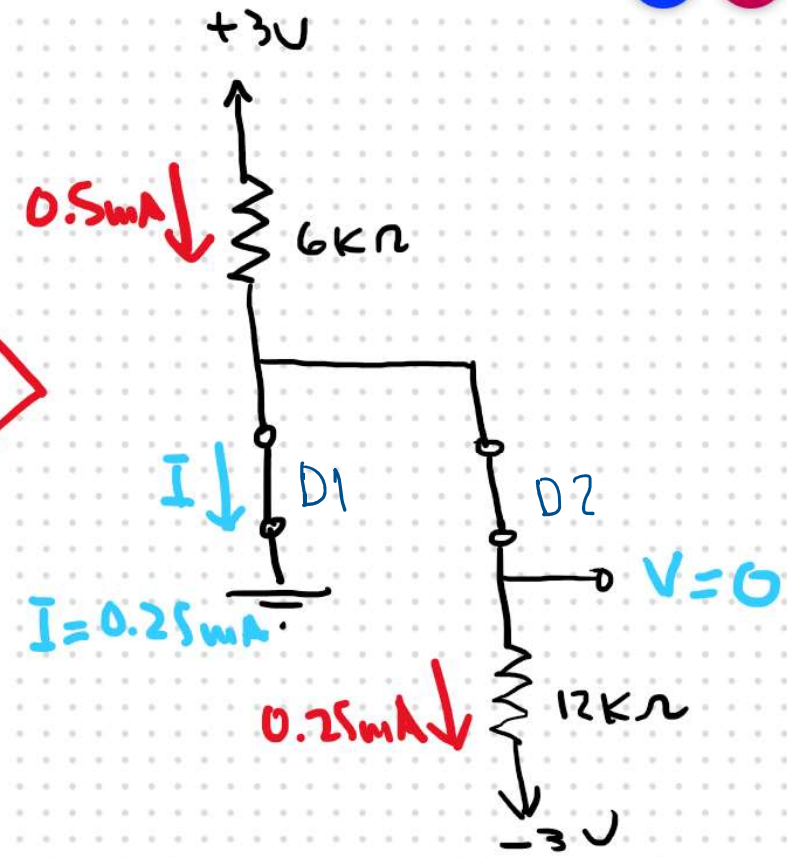
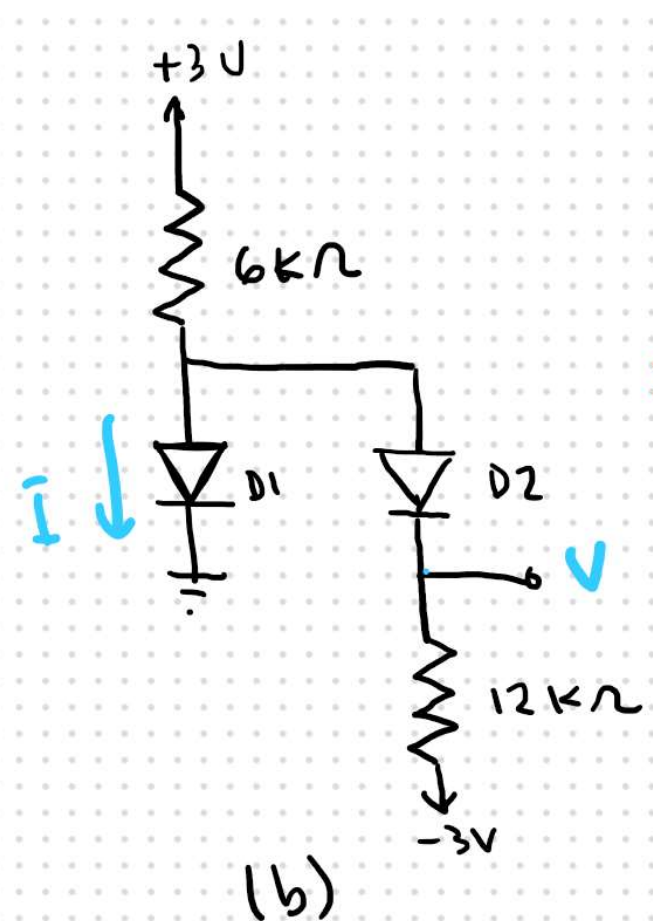
Microsoft Whiteboard toolbar containing various drawing tools: eraser, pens, highlighters, selection tools, and a text tool. A watermark "Universidad de El Salvador 2021" is visible across the toolbar.



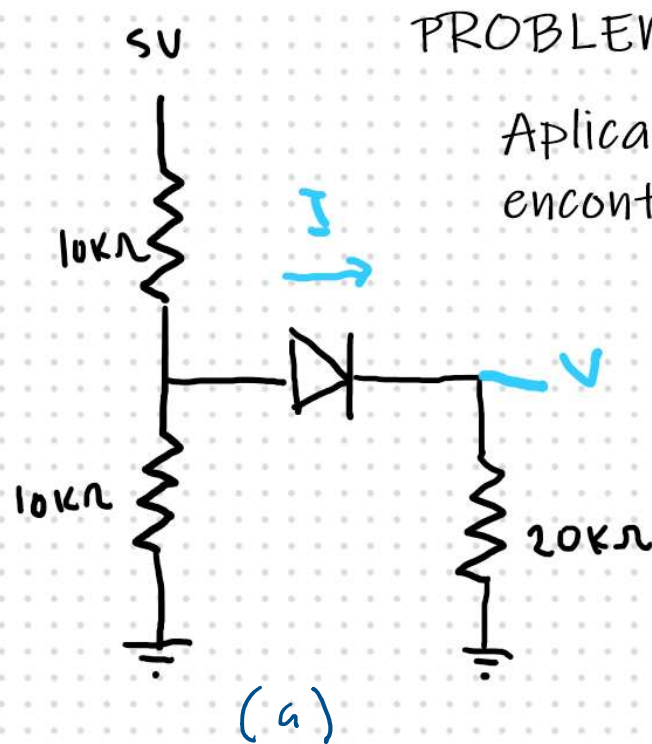


(b)





Microsoft Whiteboard toolbar with drawing tools (pens, eraser, shapes) and a watermark: Universidad de El Salvador 2021



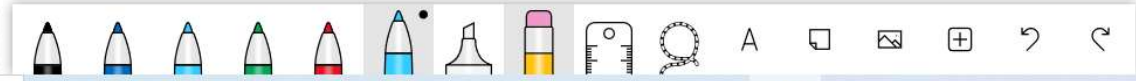
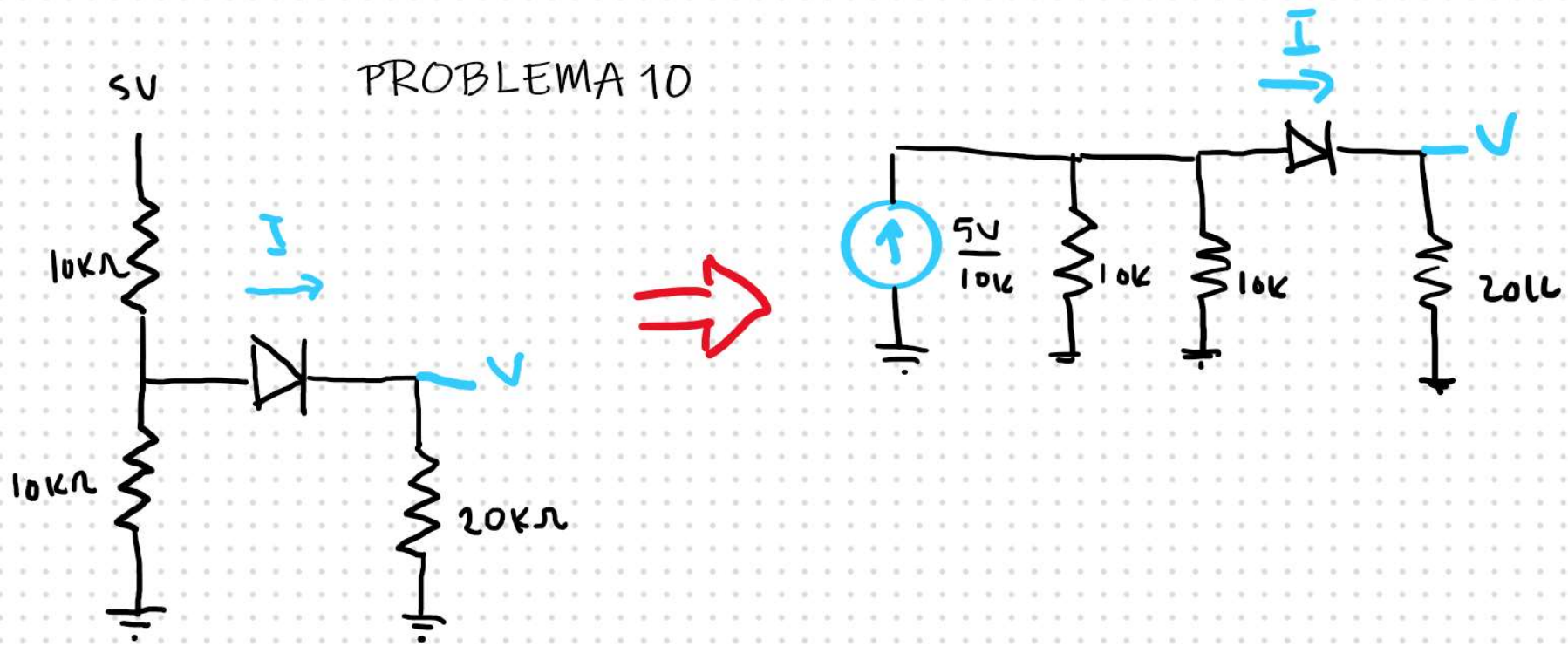
PROBLEMA 10

Aplicar Thevenin para simplificar el circuito y encontrar I y V .



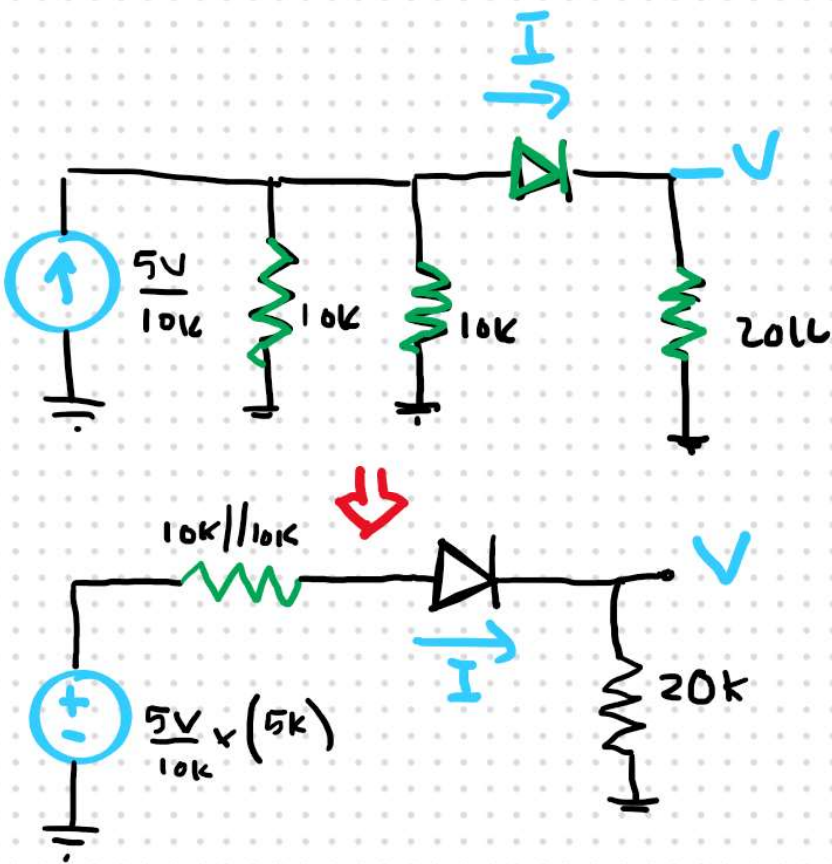
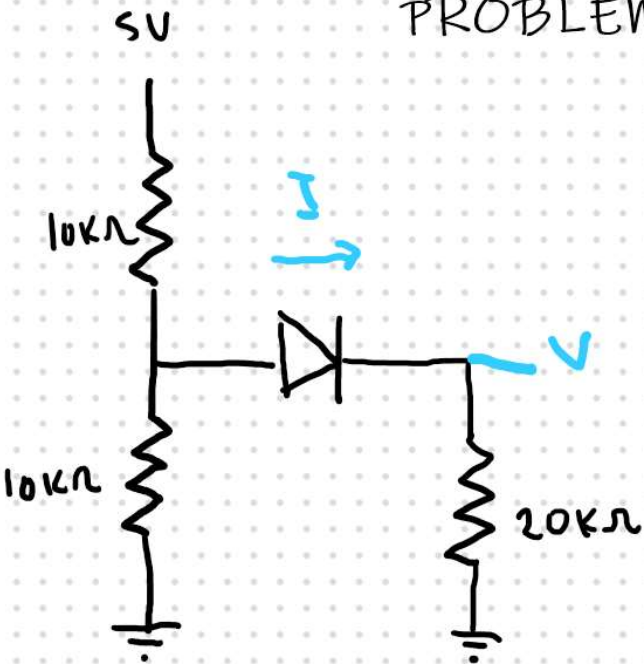


PROBLEMA 10



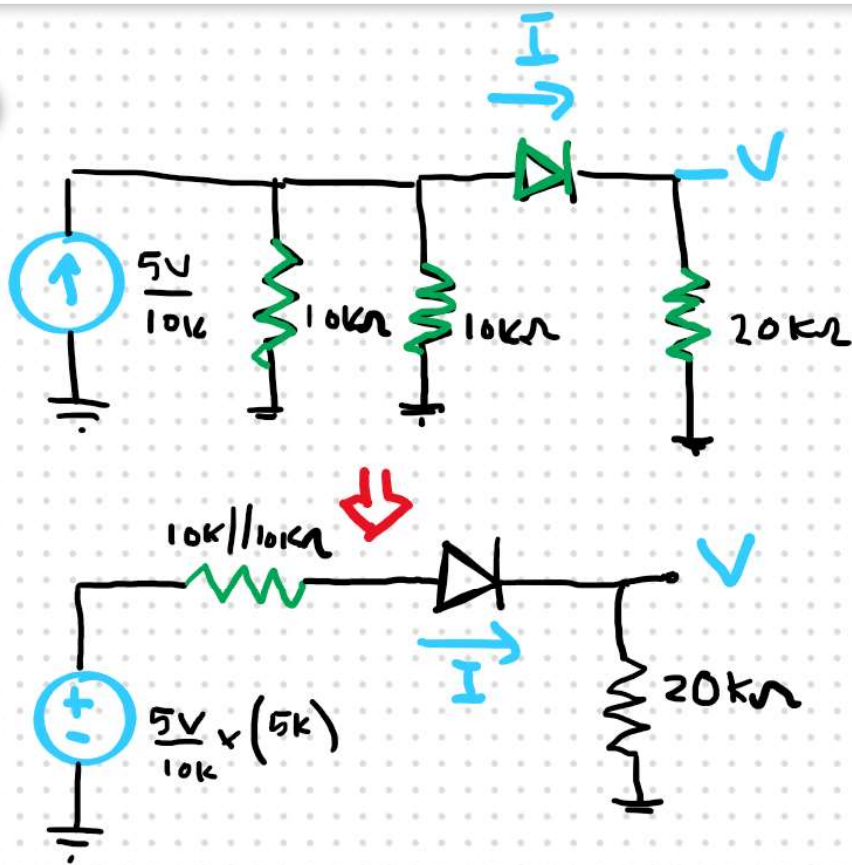


PROBLEMA 10



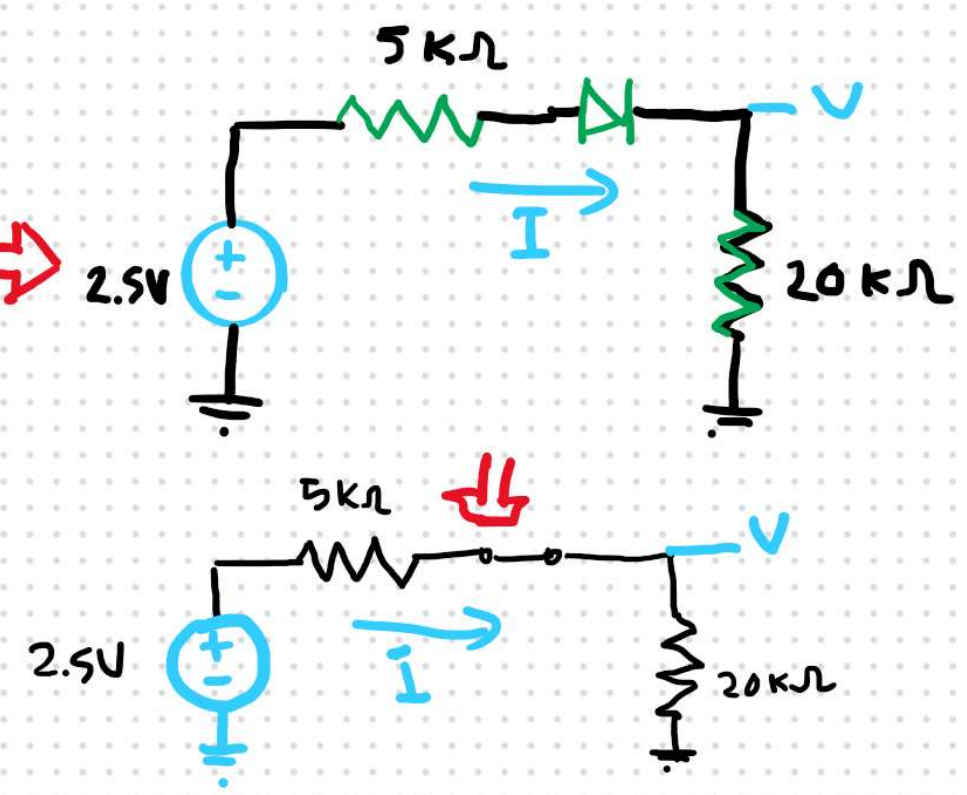
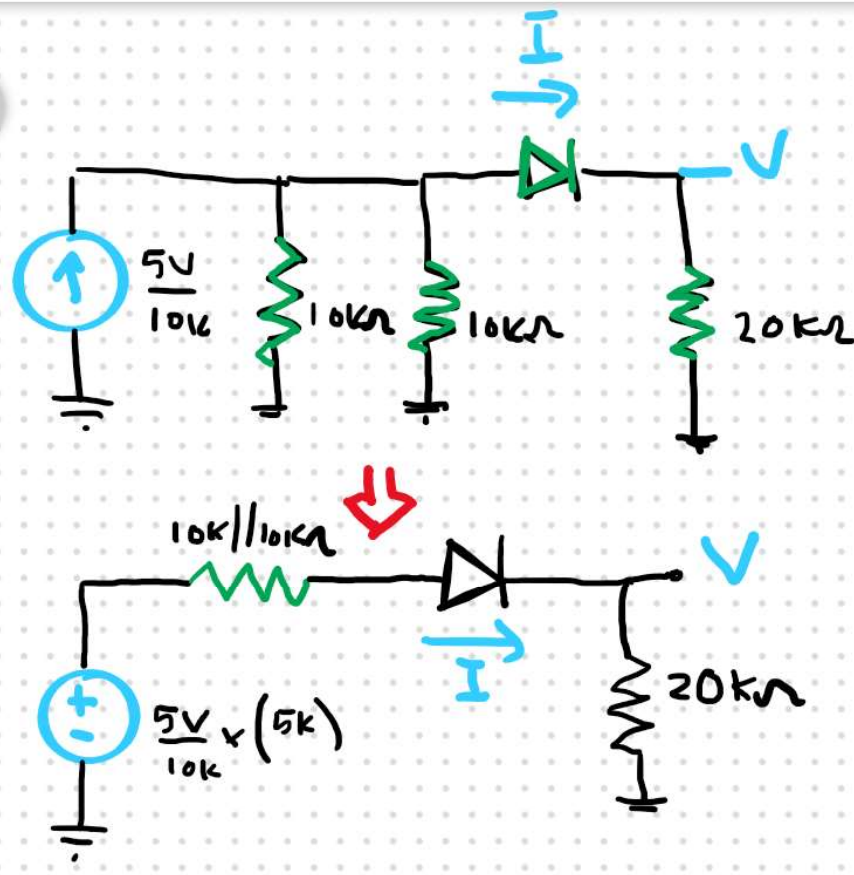
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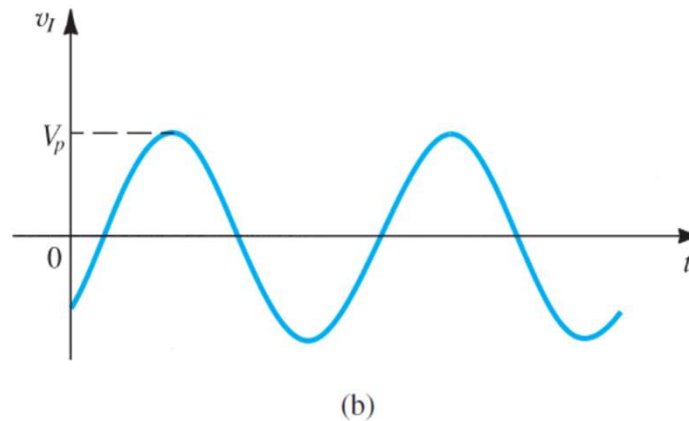
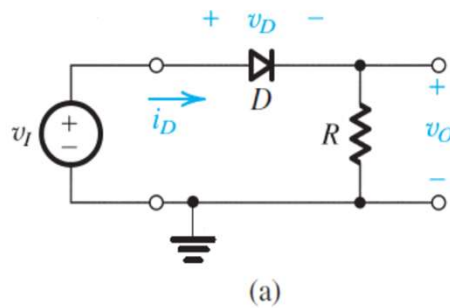


Deshacer

10

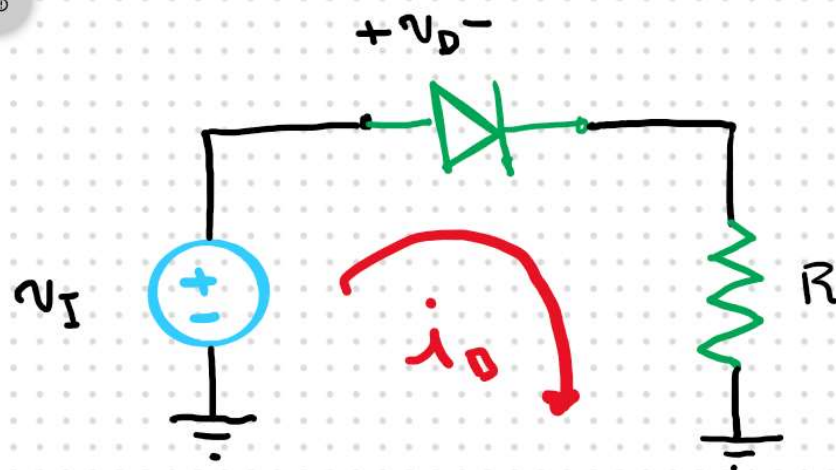


PROBLEMA 11



Fuente: Ref. [1]

Para el circuito rectificador de la figura 3(a), hagamos que la entrada tenga un valor de 120 voltios rms y asuma que el diodo es ideal. Seleccionar un valor adecuado de R , de manera que la corriente pico del diodo no exceda 40 mA. ¿Cuál es la tensión de reversa mayor que aparecerá en extremo del diodo?



$$i_D = \frac{V_m}{R} \sin \omega t$$

$$i_D = \frac{V_m \sin \omega t}{R}$$

$$I_p = \frac{V_m}{R} = \frac{120\sqrt{2}}{R}$$

$$R = \frac{169.7V}{40mA} = 4.24 k\Omega$$

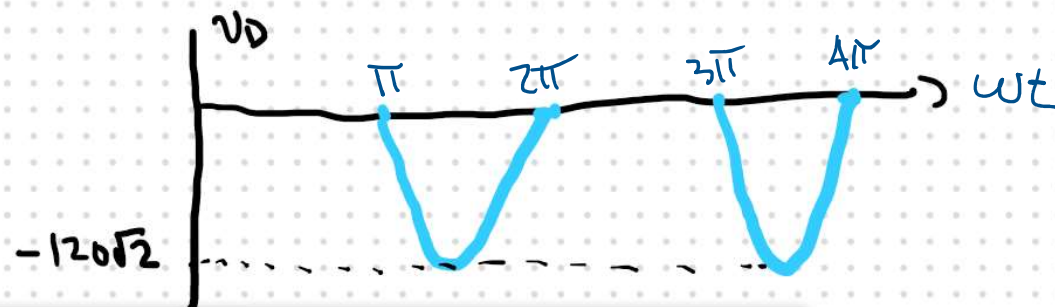
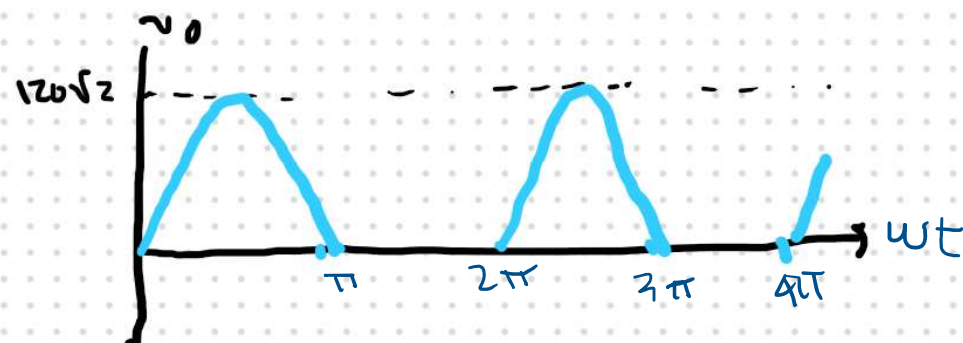
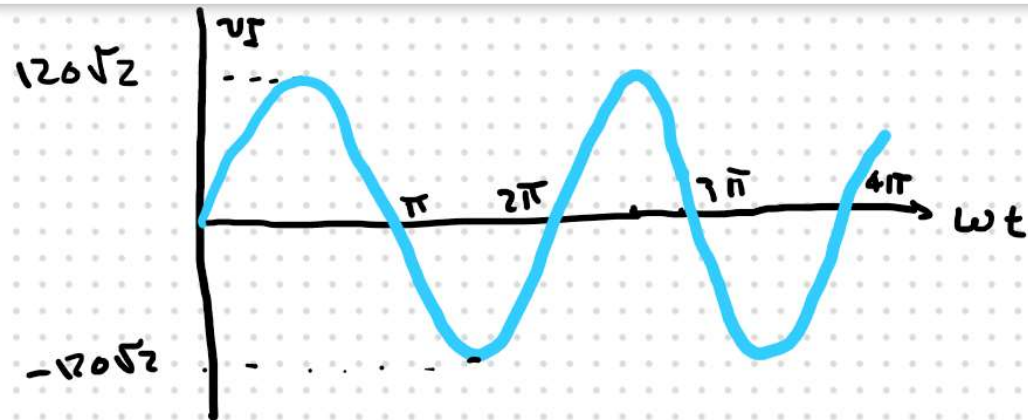




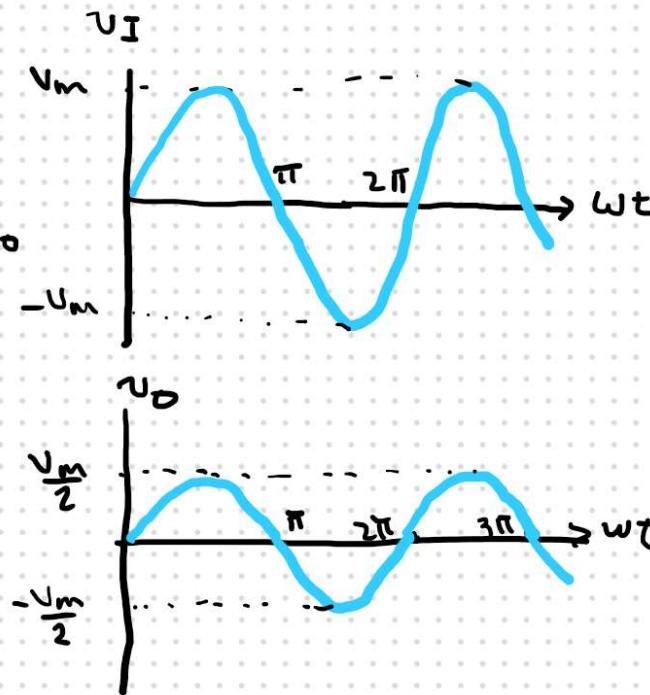
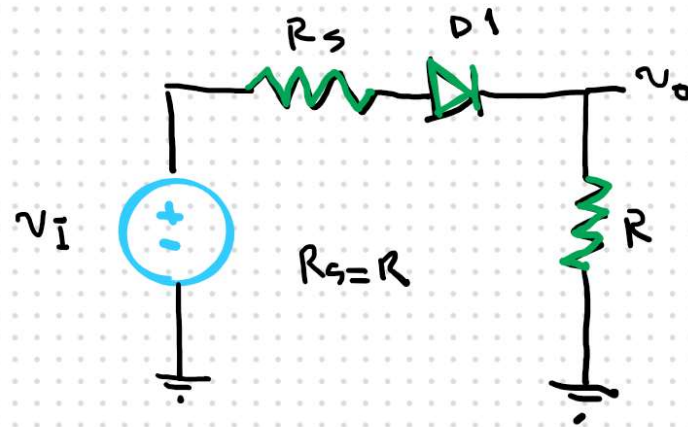
$$i_D = \frac{V_m \sin \omega t}{R}$$

$$I_p = \frac{V_m}{R} = \frac{120\sqrt{2}}{R}$$

$$R = \frac{169.7V}{40mA} = 4.24 k\Omega$$

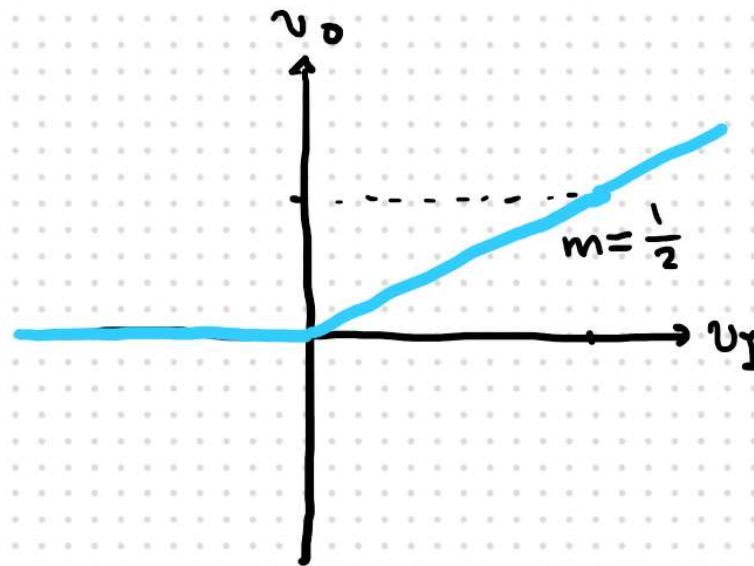


PROBLEMA 12



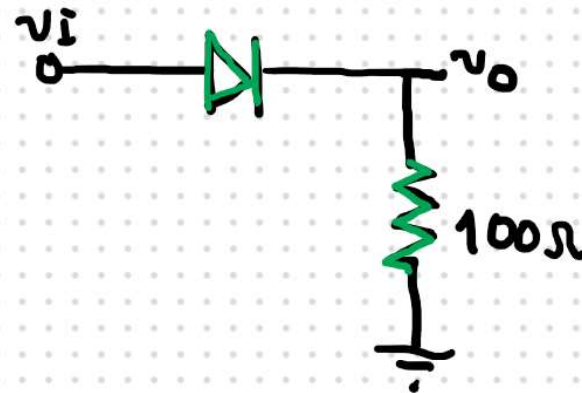
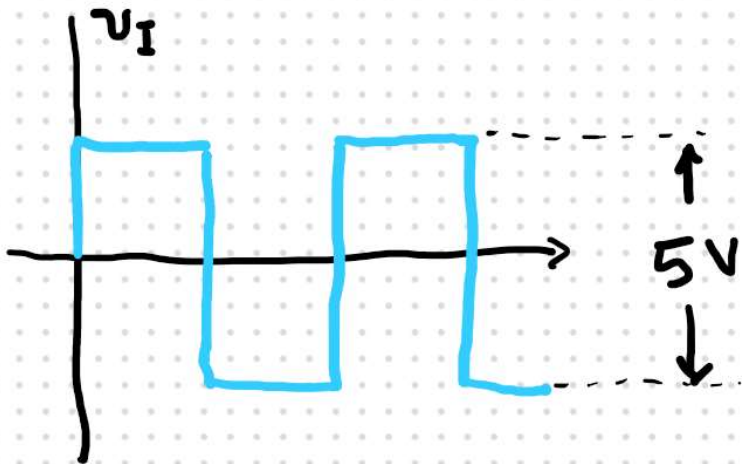
Trazar v_o y la característica de transferencia

Windows taskbar and application icons. The taskbar includes a search bar with the text "Escribe aquí para buscar". The application bar shows icons for various programs including Microsoft Edge, File Explorer, Outlook, Word, and PowerPoint. The system tray on the right shows the time as 05:44 p.m. on 04/09/2021.





PROBLEMA 13



- Trazar v_o
- ¿Tensión de salida promedio?
- ¿corriente pico del diodo?
- ¿corriente promedio del diodo?



Escribe aquí para buscar

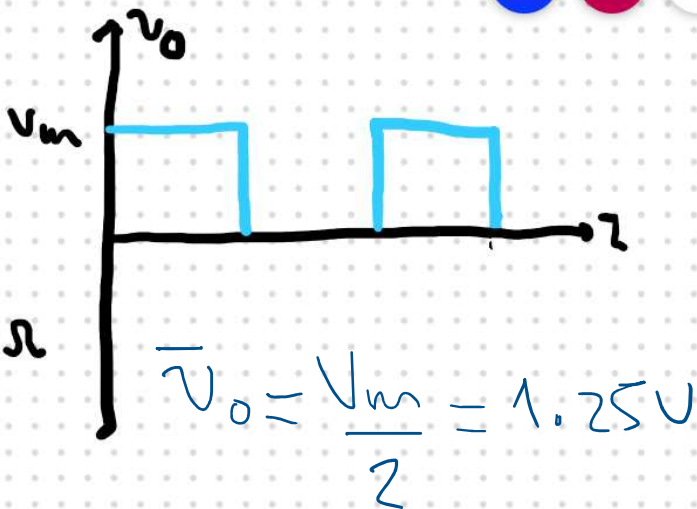
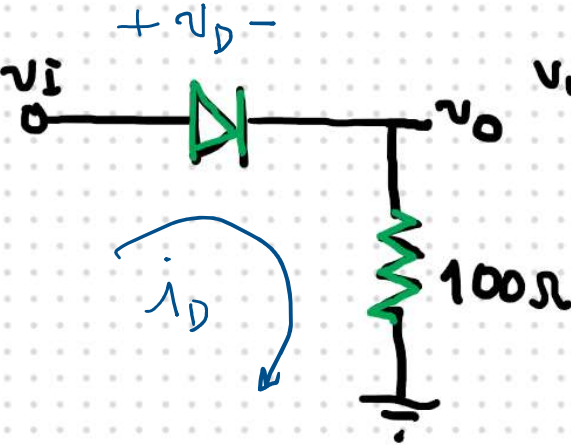
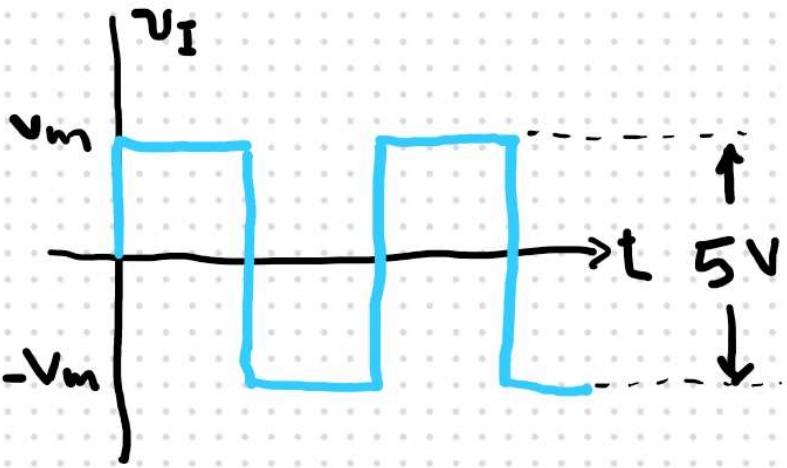


05:54 p. m.
04/09/2021



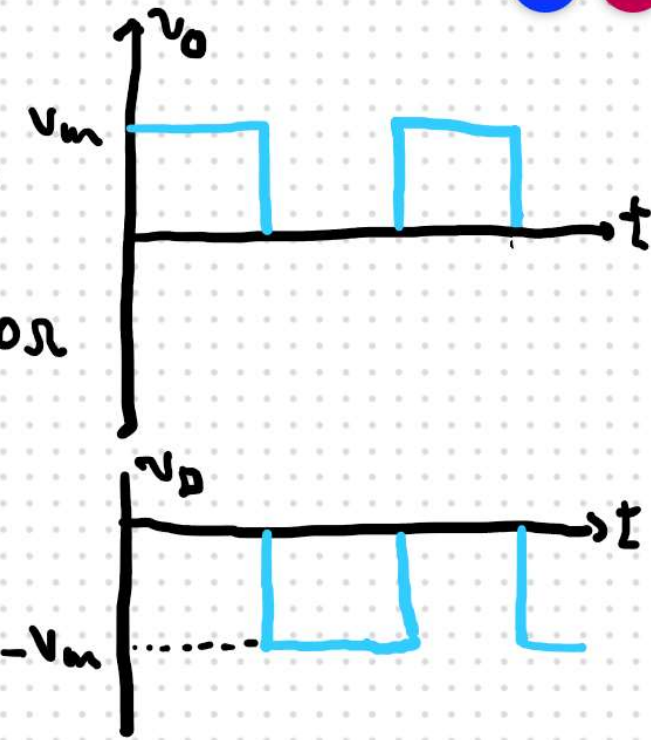
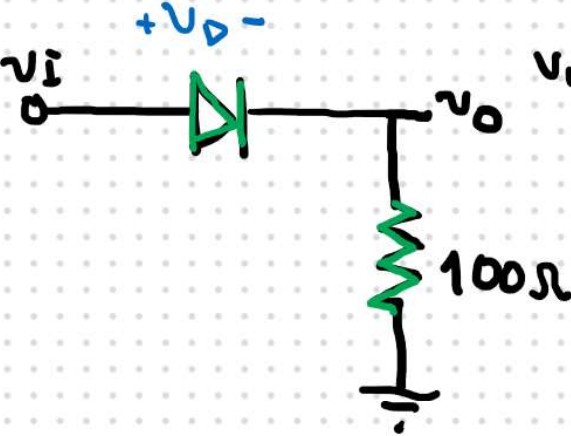
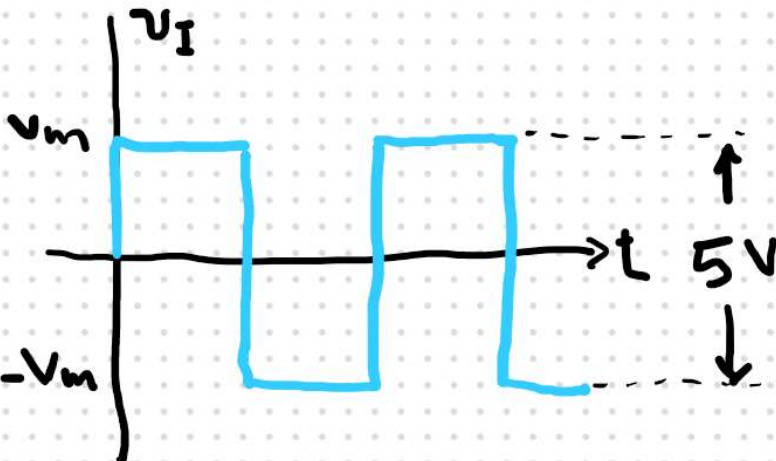


PROBLEMA 13

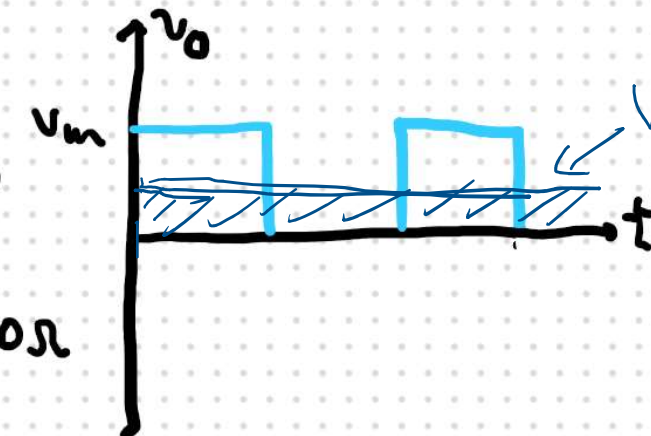
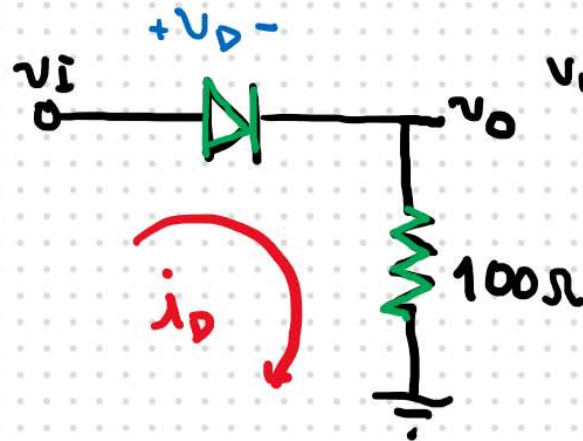
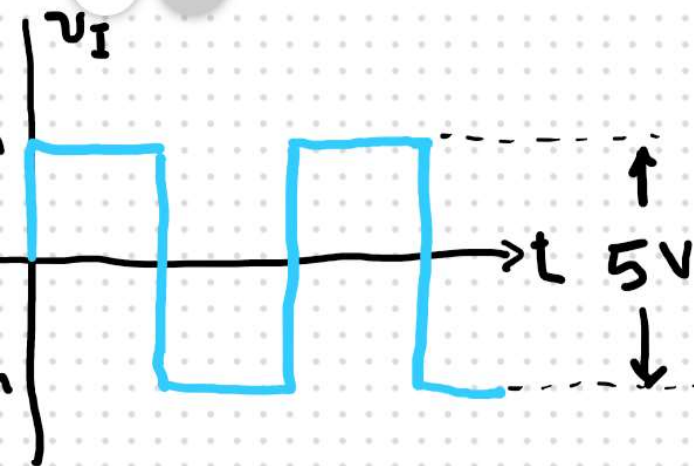




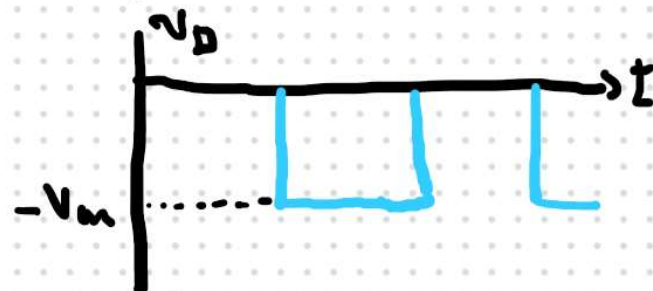
PROBLEMA 13



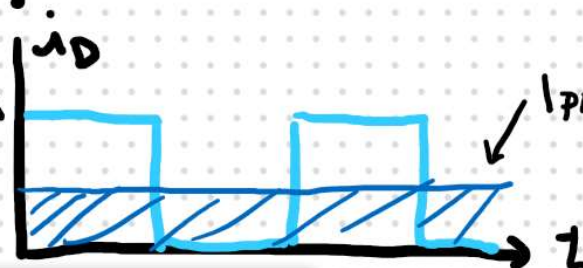
PROBLEMA 13



$$V_{PROM} = \frac{V_m}{2} = 1.25V$$



$$I_P = \frac{V_m}{R} = 25mA$$



$$I_{PROM} = \frac{I_P}{2} = 12.5mA$$



Referencias

1. Sedra & Smith. Microelectronic Circuits, 7th edition. Oxford University Press 2015.